Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	CASE-df001-2018-android
Operator:	Floris
Analysis Started:	2026-06-15 15:07:05
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\train_phantom_3\df001\2018\android
Output Directory:	C:\Users\Floris\Documents\dji_drones\train_phantom_3\df001\2018\dfdof_output\CASE-df001-2018-android
Evidence Sources:	2
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner.The populated case output directory is the authorative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
android_logical.zip	controller_android	logical	114.1 MB	2026-06-15 15:07:05
flight_logs.zip	drone_flight_storage	logical	869.5 MB	2026-06-15 15:07:08

Evidence Source Path

C:\Users\Floris\Documents\dji_drones\train_phantom_3\df001\2018\android

Cryptographic Hashes

android_logical.zip	controller_android	logical
SHA-256:	efef6b456bde504fbbbfed7e7024b170b03532e2c8bd0e948a94d49aa8210e66	
SHA-1:	c40ed6d29ec660ff226b80eabfd58ec16cf1fe16	

flight_logs.zip	drone_flight_storage	logical
SHA-256:	b0c00abba0c2deaddf3f70acf0e03b1b55b500c526b390b9d67de854e5d8b8ec	
SHA-1:	96a872d37a8b11f0d0ed78bb2da19431cdaf829b	

2. Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p2 - controller ios]: source evidence not found
2. [p3 - controller android - drone logs]: no artefacts found
3. [p3 - controller android - images]: no artefacts found
4. [p3 - controller ios]: source evidence not found
5. [p3 - drone sd]: source evidence not found
6. [p4 - controller ios]: source evidence not found
7. [p4 - drone sd]: source evidence not found
8. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY000.csv
9. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY001.csv
10. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY002.csv
11. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY003.csv
12. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY004.csv
13. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY014.csv

Tool Execution Status

Tool	Invocations	Successes	Status
datcon	22	22	[OK]
exiftool	1	1	[OK]
parser_android	1	1	[OK]
txtlogtocsvtool	1	1	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	2 / 4
Artefact Categories With Data	6 / 7
Flights With Primary Correlation	1 / 16
Tools Succeeded	4 / 4

3. Device and Account Identification

Field	Value	Confidence	Sources
Installed DJI App(s)	dji.pilot	[K] Controller-only	1
Drone Serial Number	03Z1389682	[K] Controller-only	2
Drone Model	P3 Standard P3S	[!] Inconsistent	17
DJI Application Version	3.1.38	[K] Controller-only	1
Product Type	2	[K] Controller-only	1
Firmware Version	1.9.20	[K] Controller-only	1
Last Known Latitude	39.96118719	[K] Controller-only	1
Last Known Longitude	-106.21645676	[K] Controller-only	1
Account UID	879825329191059456	[K] Controller-only	1
Device UUID	2b47db92-c893-4aae-892b-433e63d87543	[K] Controller-only	1

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status:	Unmatched (no drone log corroboration)
Drone Log SHA-256:	`6330a8eedcd3f8ff7a6ba78b4355738d4056f98b4bec08831507aa2568c34c62`
Drone Log Name:	FLY005.csv
Start:	2018-06-07T15:49:55.557000+00:00
End:	2018-06-07T15:54:24.160000+00:00
Duration:	268.6 s

Flight Metrics

Peak Height:	-
Distance Travelled:	-
Video Recording:	No
Photo Capture:	No

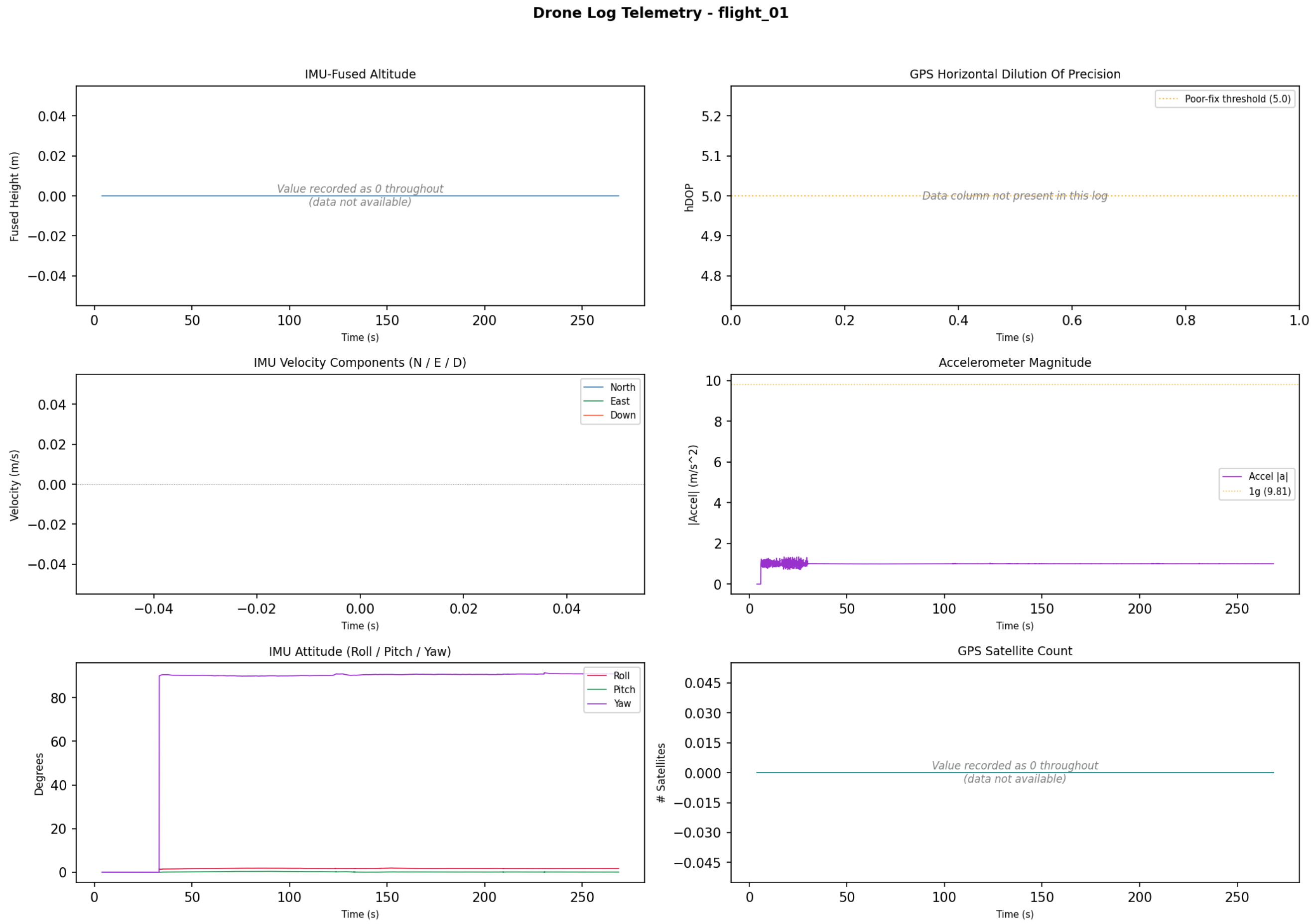
Fly Modes Observed (Chronological)

1. 0

Key Events

Timestamp (UTC)	Source	Event	Details
2018-06-07 15:50:47	drone_flight_storage:dl	Log started	P3S
2018-06-07 15:54:24	drone_flight_storage:dl	Log ended	
3236-01-12 23:59:48	drone_flight_storage:dl	Reached peak height	-0.0 m

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 02

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `65ca3eb3e94b3293c8627f682f92945867acbb5944751cbcc13c7d29659d8043`
Drone Log Name: FLY006.csv
Start: 2018-06-07T16:41:32.117000+00:00
End: 2018-06-07T16:43:53.300000+00:00
Duration: 141.2 s

Flight Metrics

Peak Height: -
Distance Travelled: 0.01228646116311485 m
Video Recording: No
Photo Capture: No

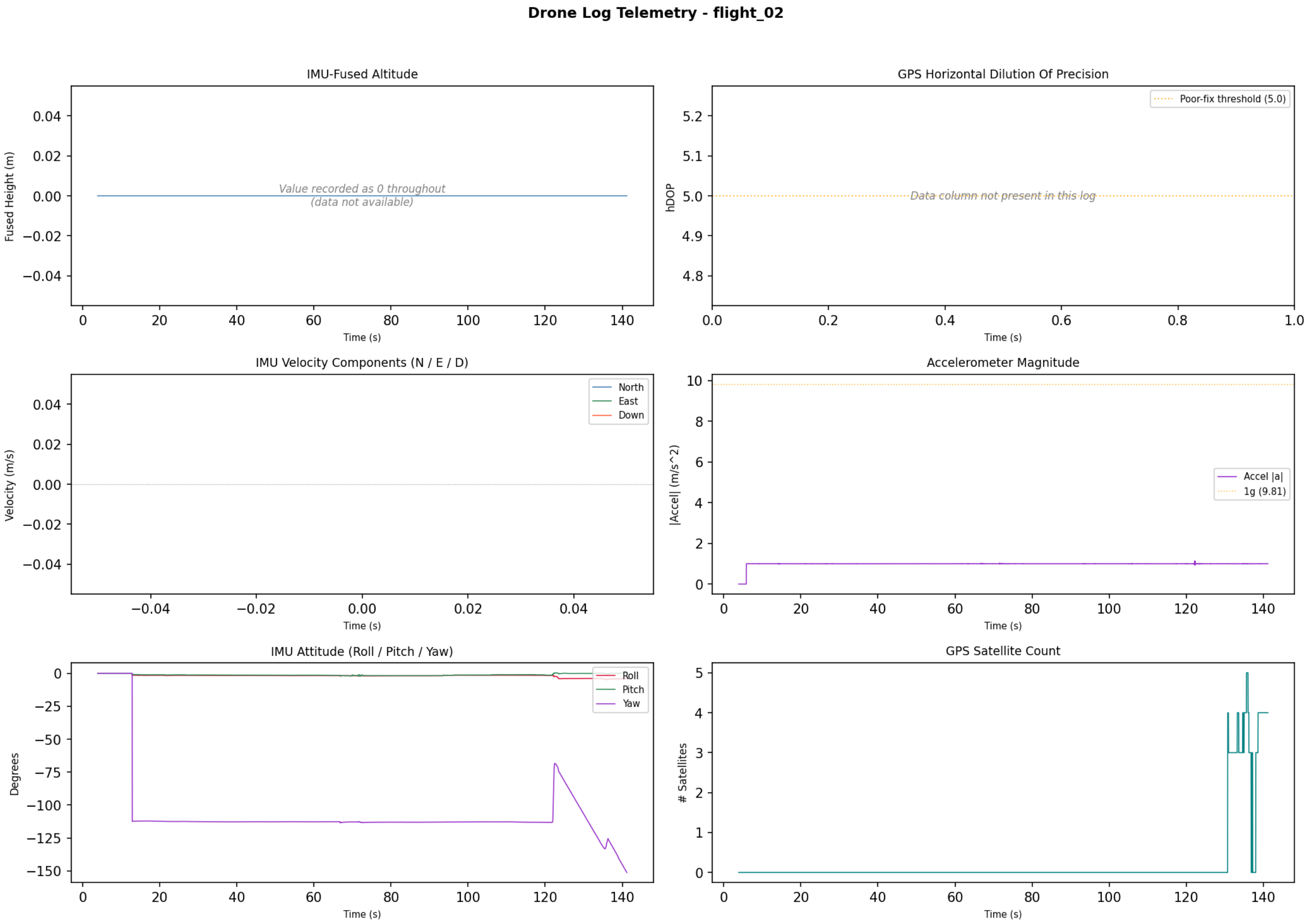
Fly Modes Observed (Chronological)

1. 0

Key Events

Timestamp (UTC)	Source	Event	Details
2018-06-07 16:41:36	drone_flight_storage:dl	Log started	P3S
2018-06-07 16:41:36	drone_flight_storage:dl	Reached peak height	-0.0 m
2018-06-07 16:43:56	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 03

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `ae702aa8ef96dbfe2763f478fe48ebfefff50092f1a3e0976cb4861899e0edcd`
Drone Log Name: FLY007.csv
Start: 2018-06-07T16:48:35.118000+00:00
End: 2018-06-07T16:50:51.300000+00:00
Duration: 136.2 s

Flight Metrics

Peak Height: -
Distance Travelled: 58.783664825212966 m
Video Recording: No
Photo Capture: No

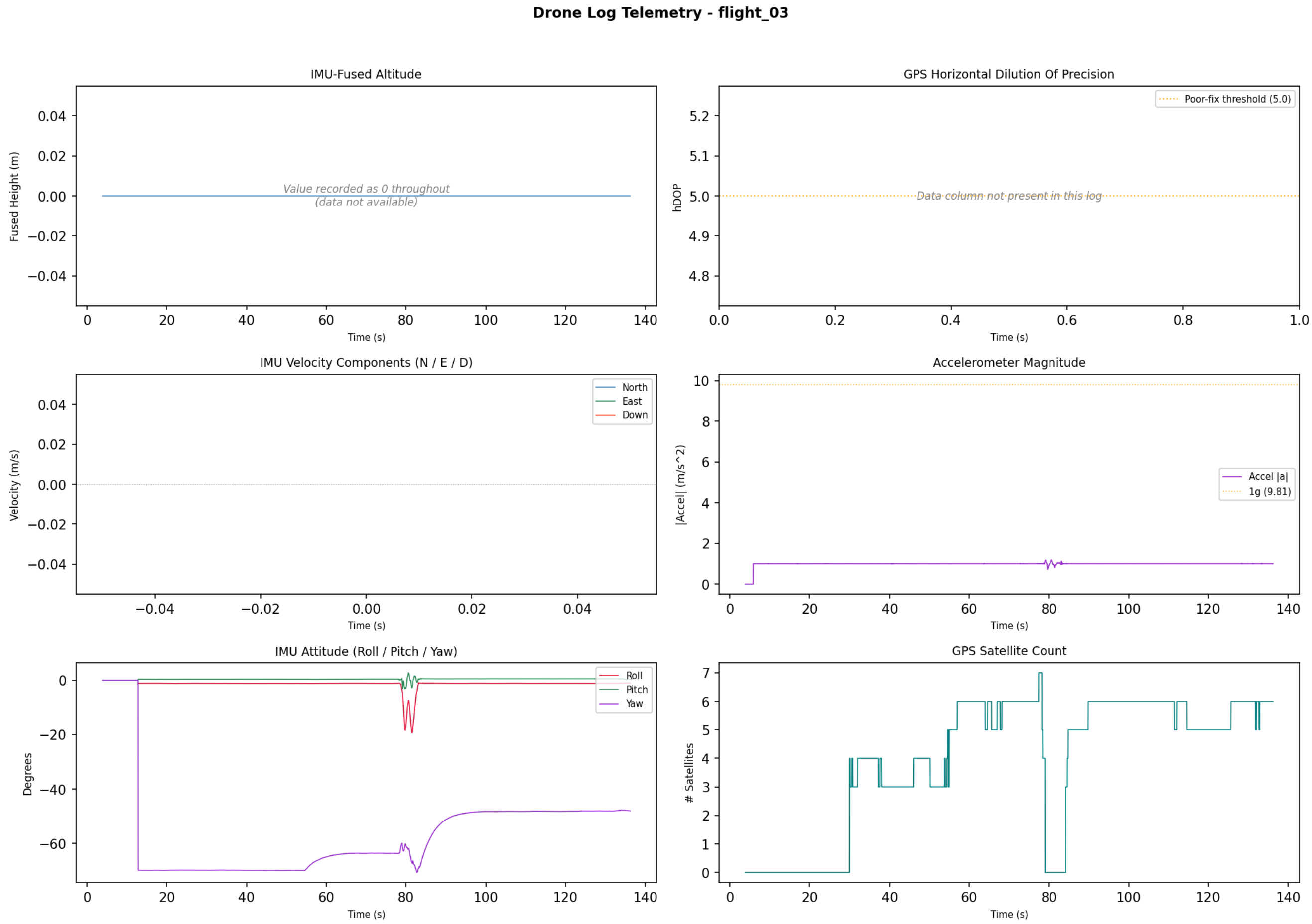
Fly Modes Observed (Chronological)

1. 0

Key Events

Timestamp (UTC)	Source	Event	Details
2018-06-07 16:48:39	drone_flight_storage:dl	Log started	P3S
2018-06-07 16:48:39	drone_flight_storage:dl	Reached peak height	-0.0 m
2018-06-07 16:50:54	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 04

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `a27ad6b90d8e3d2ce3f4715a93b866948df4f66ec7faf8fa74a14ab21fc994ff`
Drone Log Name: FLY009.csv
Start: 2018-06-16T23:59:46.463000+00:00
End: 2018-06-17T00:05:48.015000+00:00
Duration: 361.6 s

Flight Metrics

Peak Height: 0.09117607116368376 m
Distance Travelled: 4.0208613717078165 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

1. 0

Key Events

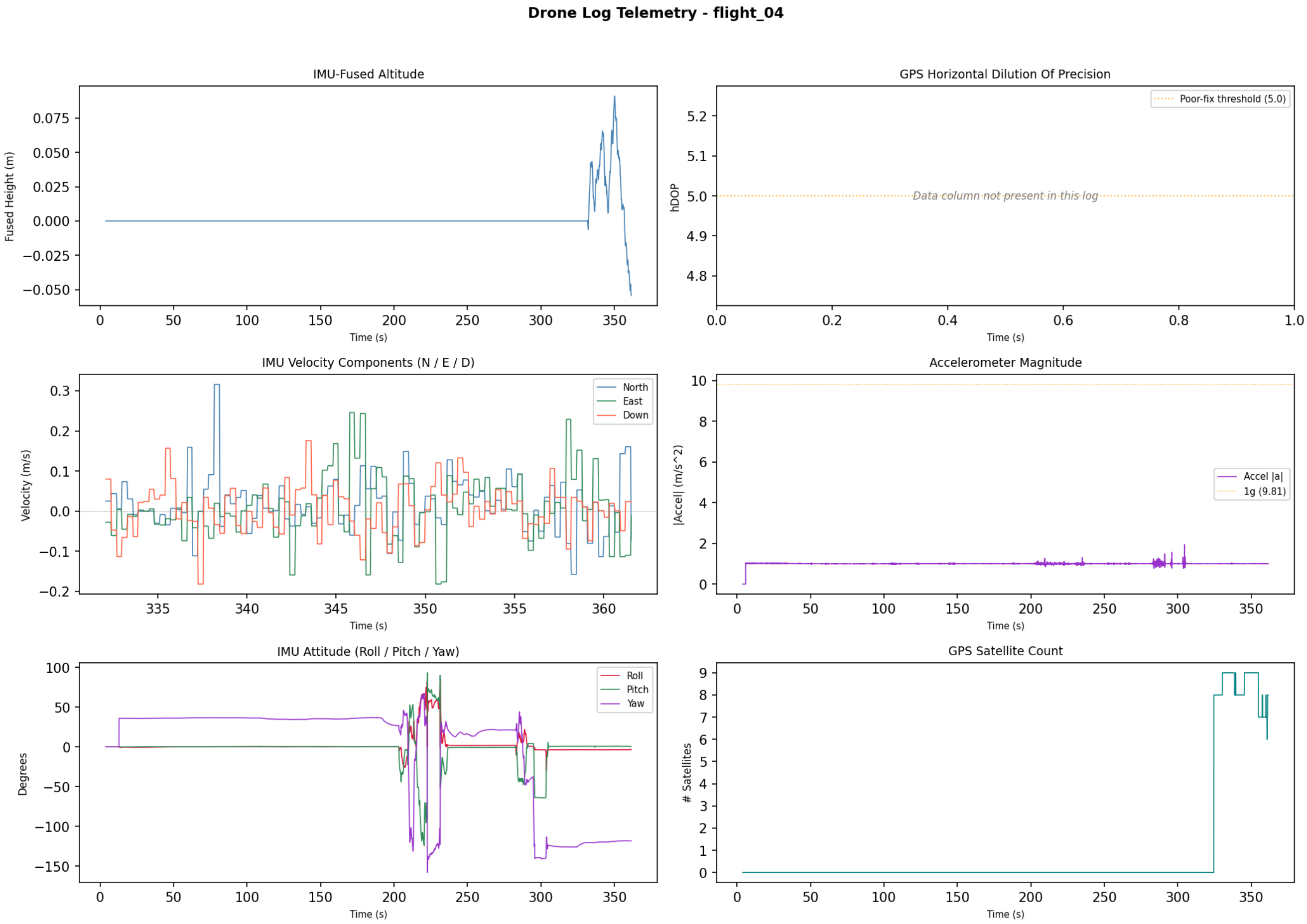
Timestamp (UTC)	Source	Event	Details
2018-06-17 00:04:36	drone_flight_storage:dl	Log started	P3S
2018-06-20 14:21:22	drone_flight_storage:dl	Reached peak height	0.1 m
2018-06-20 14:21:34	drone_flight_storage:dl	Log ended	

Correlated Artefacts

Possibly correlated (indirect indicators):

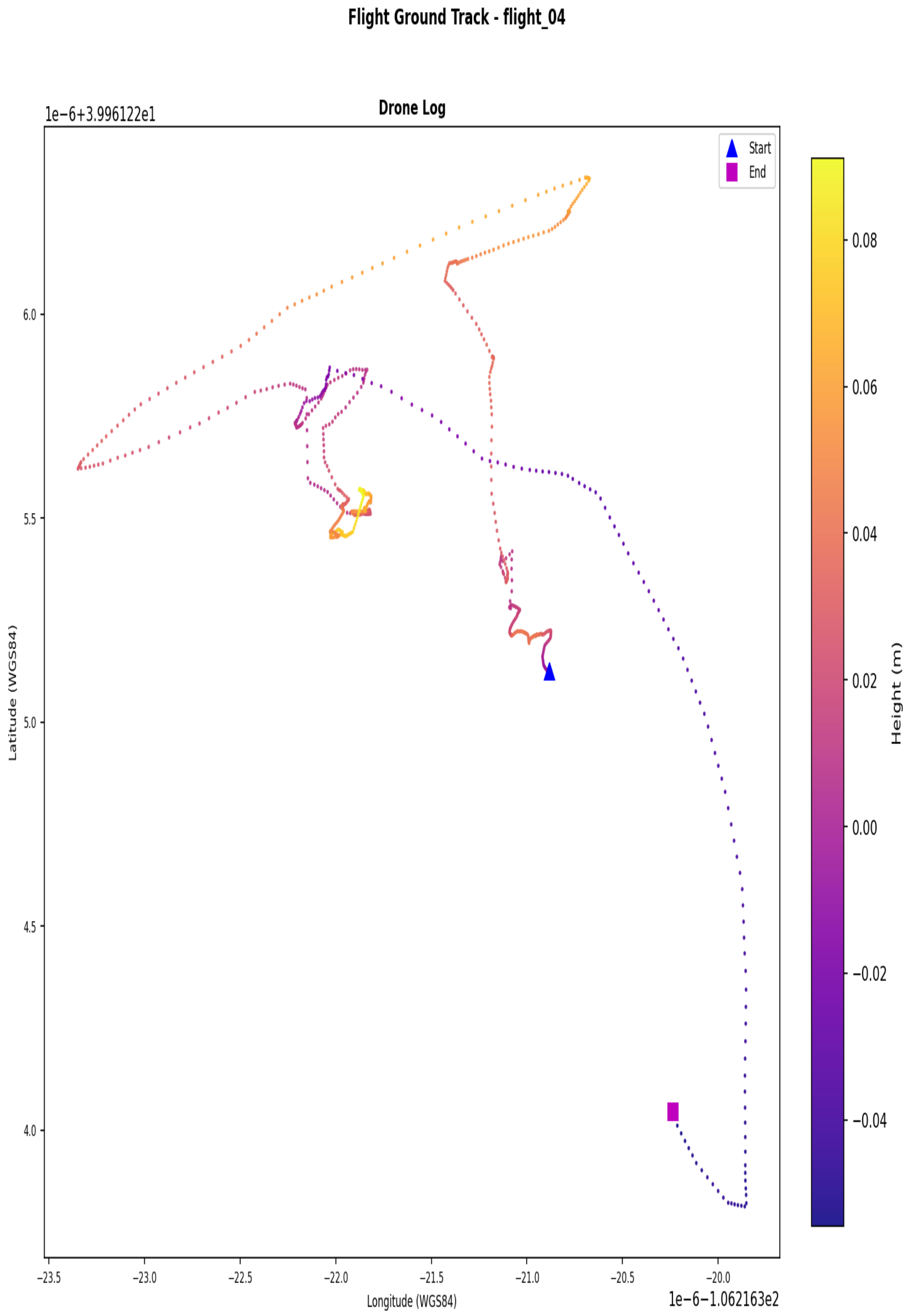
controller_android:videos	GPS within flight bounding box (lat 39.9623, lon -106.2198)
p5:fe28dc748339f681a0b0ec8fe9182f15b778ec23f27bf80d7998ddd6125961f7	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY009.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).
A KML file (FLY009.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 05

Correlation and Flight Window

Status:	Unmatched (no drone log corroboration)
Drone Log SHA-256:	`72f271c20306ff986e07eb0a8e5de564dd77339a67955cc17146c06d43ed6eb2`
Drone Log Name:	FLY008.csv
Start:	2018-06-19T18:59:03.770000+00:00
End:	2018-06-19T19:23:38.760000+00:00
Duration:	1475.0 s

Flight Metrics

Peak Height:	39.45516735027209 m
Distance Travelled:	1710.0320023601398 m
Video Recording:	No
Photo Capture:	No

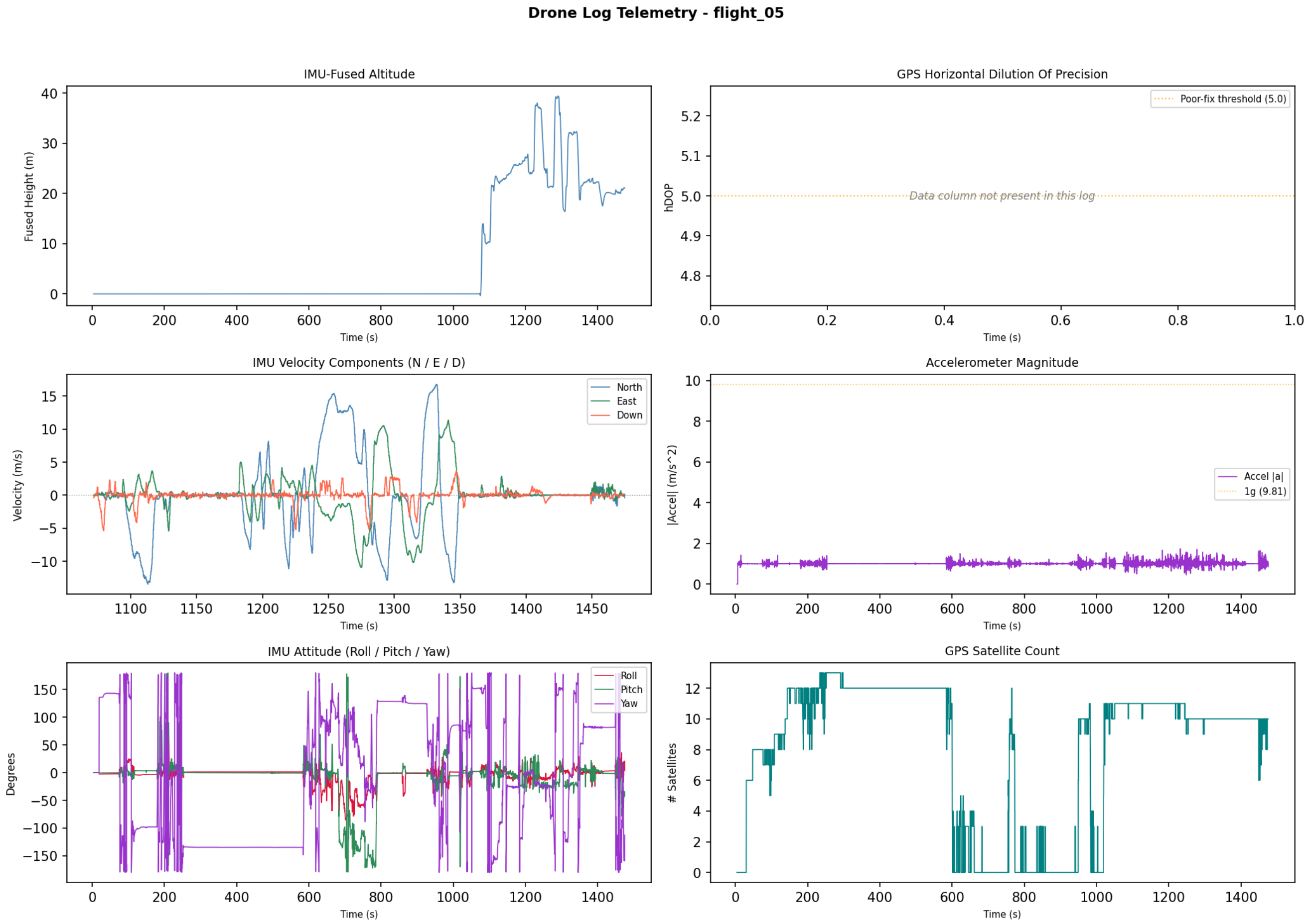
Fly Modes Observed (Chronological)

1. 1
2. 2
3. 3
4. 0

Key Events

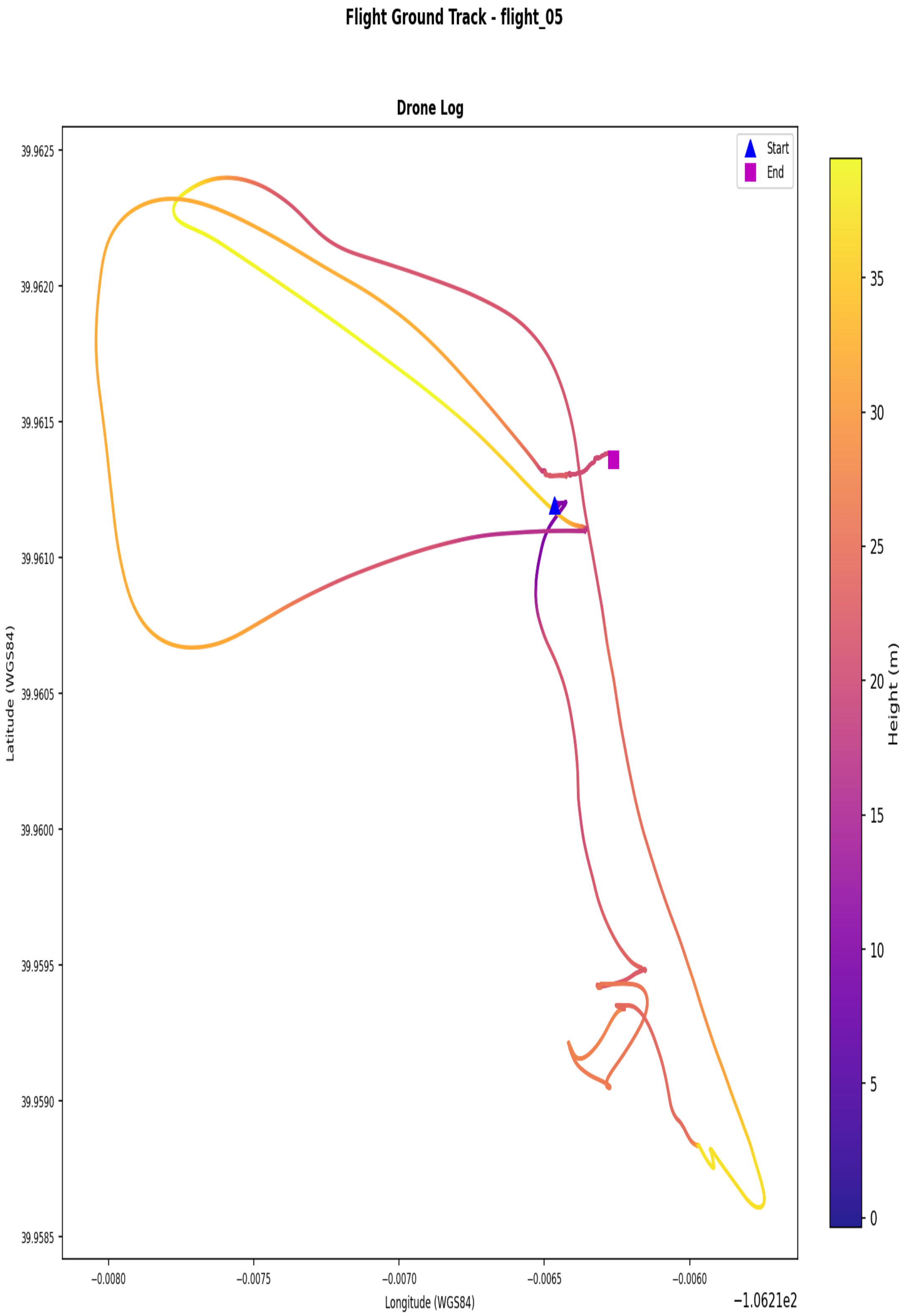
Timestamp (UTC)	Source	Event	Details
2018-06-19 18:59:17	drone_flight_storage:dl	Log started	P3S
2018-06-19 19:20:30	drone_flight_storage:dl	Reached peak height	39.5 m
2018-06-19 19:23:35	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY008.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY008.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 06

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `78fbdc9fab8a9fb5c2eeeebb7ce94576158c456fcda5520ab06fb9206d9f5af1`
Drone Log Name: FLY010.csv
Start: 2018-06-20T14:40:43.103000+00:00
End: 2018-06-20T14:40:54.105000+00:00
Duration: 11.0 s

Flight Metrics

Peak Height: -
Distance Travelled: -
Video Recording: No
Photo Capture: No

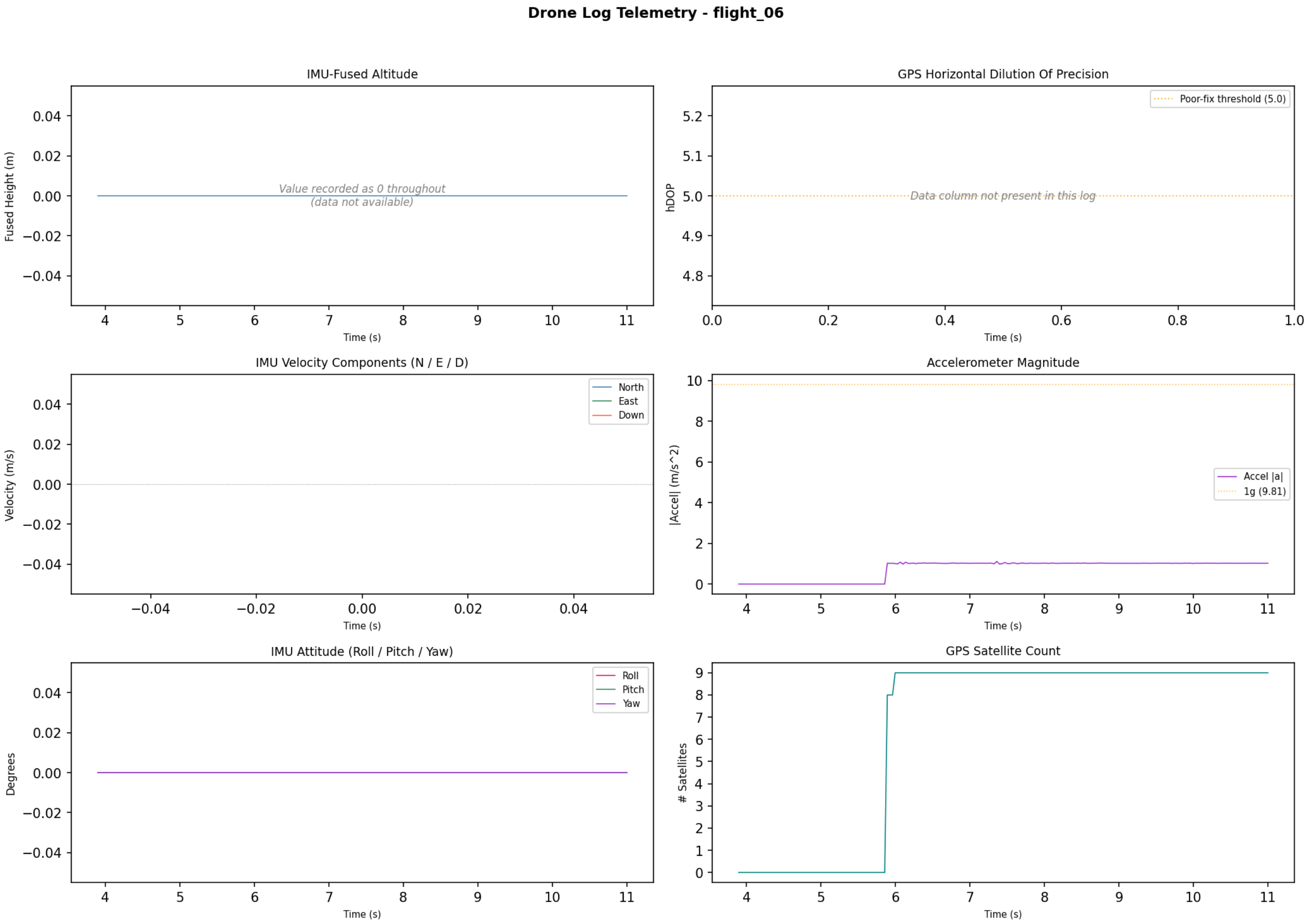
Fly Modes Observed (Chronological)

1. 0

Key Events

Timestamp (UTC)	Source	Event	Details
2018-06-20 14:40:47	drone_flight_storage:dl	Log started	P3S
2018-06-20 14:40:47	drone_flight_storage:dl	Reached peak height	-0.0 m
2018-06-20 14:40:56	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 07

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `bdff755e0f19d86a907baefc21a0ccb0bf6c1df389c378589c2853f5b3c3be0f`
Drone Log Name: FLY011.csv
Start: 2018-06-20T14:58:54.075000+00:00
End: 2018-06-20T15:00:26.280000+00:00
Duration: 92.2 s

Flight Metrics

Peak Height: 7.154927647096823 m
Distance Travelled: 11.007765231291225 m
Video Recording: No
Photo Capture: No

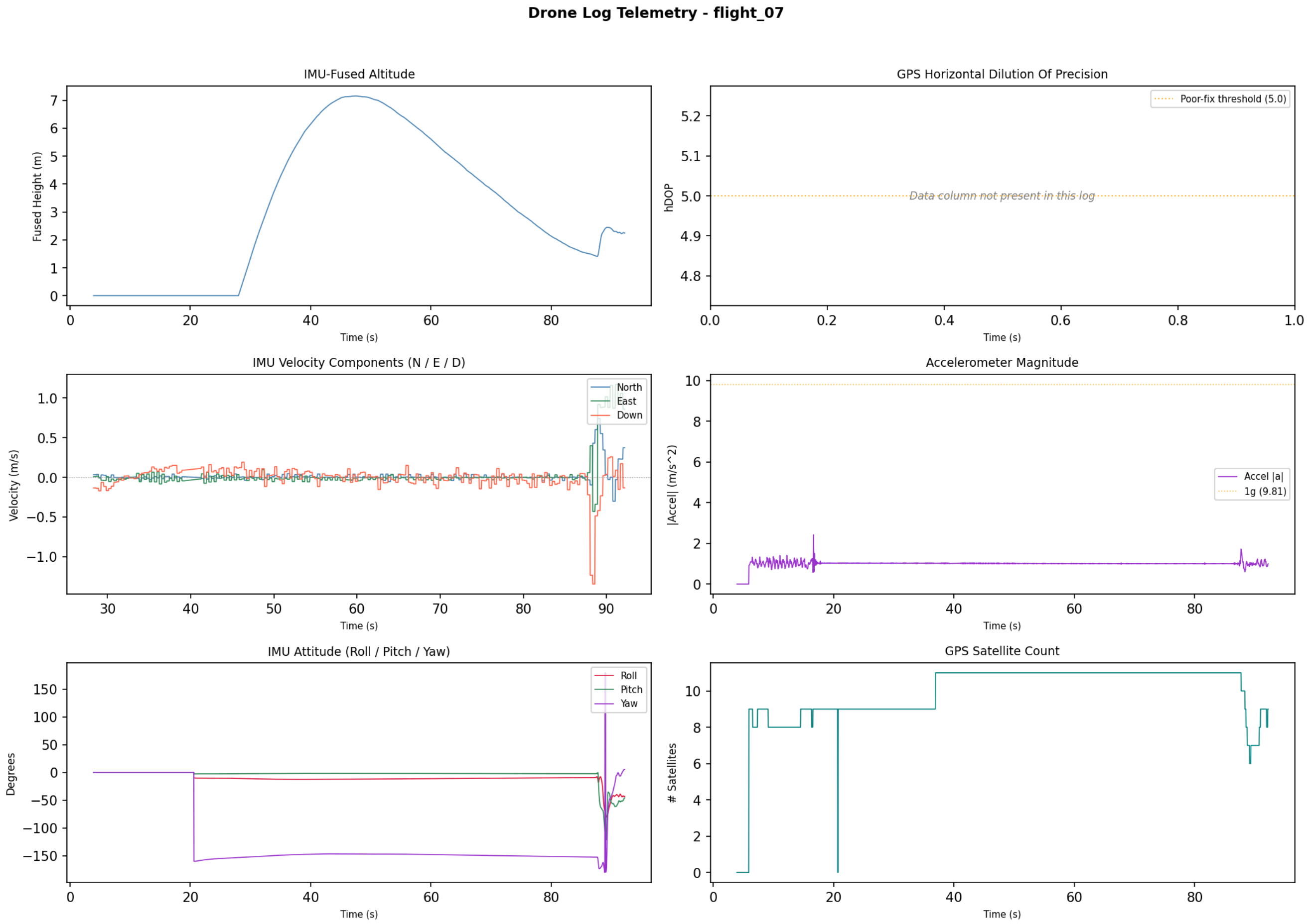
Fly Modes Observed (Chronological)

1. 0

Key Events

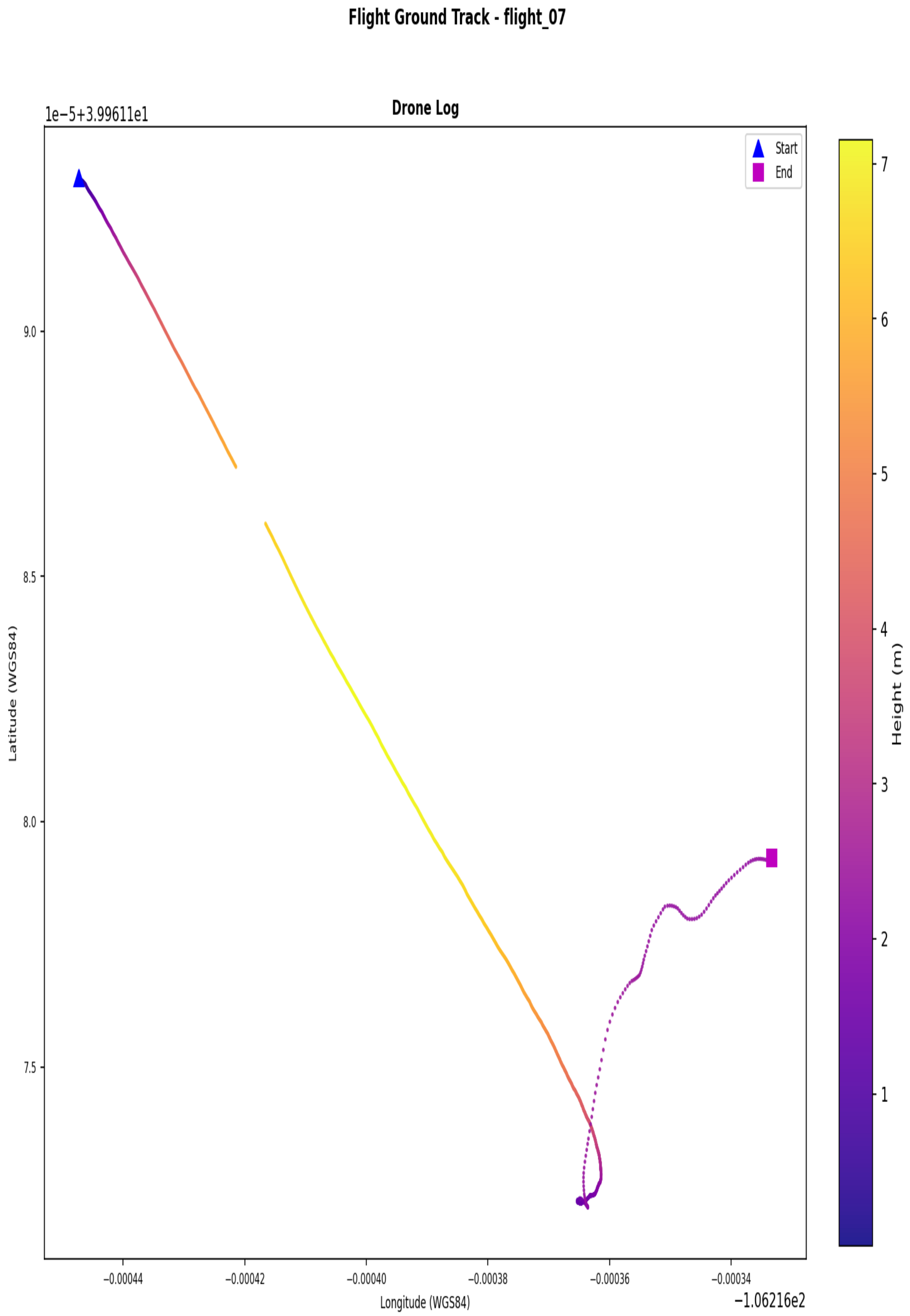
Timestamp (UTC)	Source	Event	Details
2018-06-20 14:58:58	drone_flight_storage:dl	Log started	P3S
2018-06-20 14:59:43	drone_flight_storage:dl	Reached peak height	7.2 m
2018-06-20 15:00:28	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY011.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).
A KML file (FLY011.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 08

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `c96fa8c2fb3bc8480e67f51c197f8c1d32c7c4953dae24d371e09b2823ddbc20`
Drone Log Name: FLY012.csv
Start: 2018-06-20T15:01:11.105000+00:00
End: 2018-06-20T15:03:13.800000+00:00
Duration: 122.7 s

Flight Metrics

Peak Height: 1.3026932382708851 m
Distance Travelled: 4.927617715712613 m
Video Recording: No
Photo Capture: No

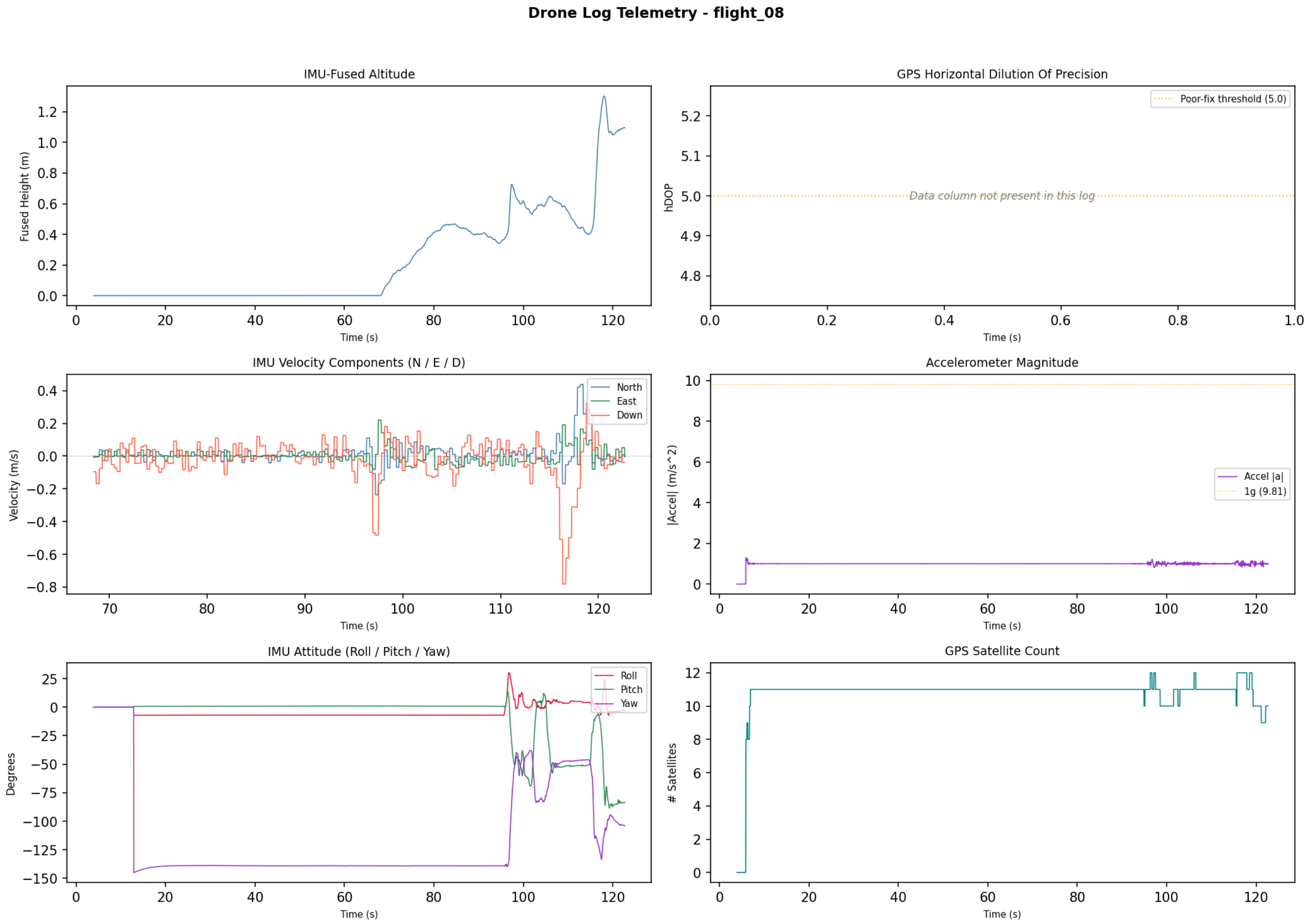
Fly Modes Observed (Chronological)

1. 0

Key Events

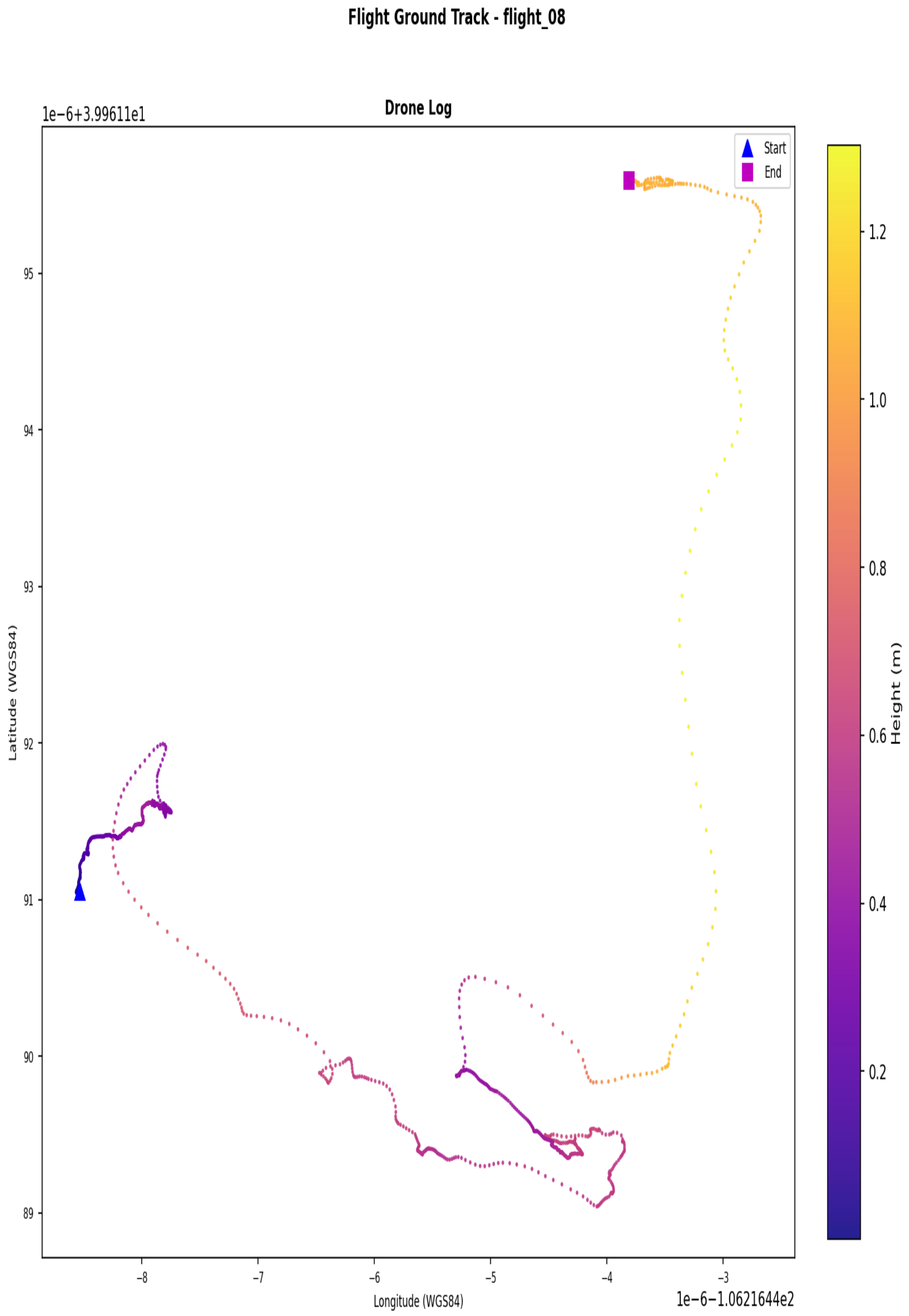
Timestamp (UTC)	Source	Event	Details
2018-06-20 15:01:15	drone_flight_storage:dl	Log started	P3S
2018-06-20 15:03:11	drone_flight_storage:dl	Reached peak height	1.3 m
2018-06-20 15:03:16	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY012.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY012.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 09

Correlation and Flight Window

Status:	Matched (primary GPS+temporal)
Confidence:	high
GPS Overlap:	228.2 s
Median GPS Distance:	0.78 m
Overlap:	100%
GPS Match Rate:	100%
Flight Record SHA-256:	0aaa0febea9ba19ef91afbf4ae1fef64703f104e86269985f6ce94669ffc1a06
Flight Record Name:	DJIFlightRecord_2018-06-20_[09-04-19].csv
Start:	2018-06-20T15:04:19.862000+00:00
End:	2018-06-20T15:08:08.057000+00:00
Duration:	228.2 s
Drone Log SHA-256:	b4e674e670d3e40d779bf7b0856d21225202313e220e194b2a4d853bbef740ca
Drone Log Name:	FLY013.csv
Start:	2018-06-20T15:03:38.065000+00:00
End:	2018-06-20T15:08:35.560000+00:00
Duration:	297.5 s

Flight Metrics

Peak Height:	53.9 m
Distance Travelled:	1693.05 m
Video Recording:	Yes
Photo Capture:	No
Photos Taken:	0
Recording Duration:	0.0 s
Record Complete:	Yes

Fly Modes Observed (Chronological)

1. 0
2. GPS_Atti
3. 2
4. AutoTakeoff
5. 1
6. 3

Key Events

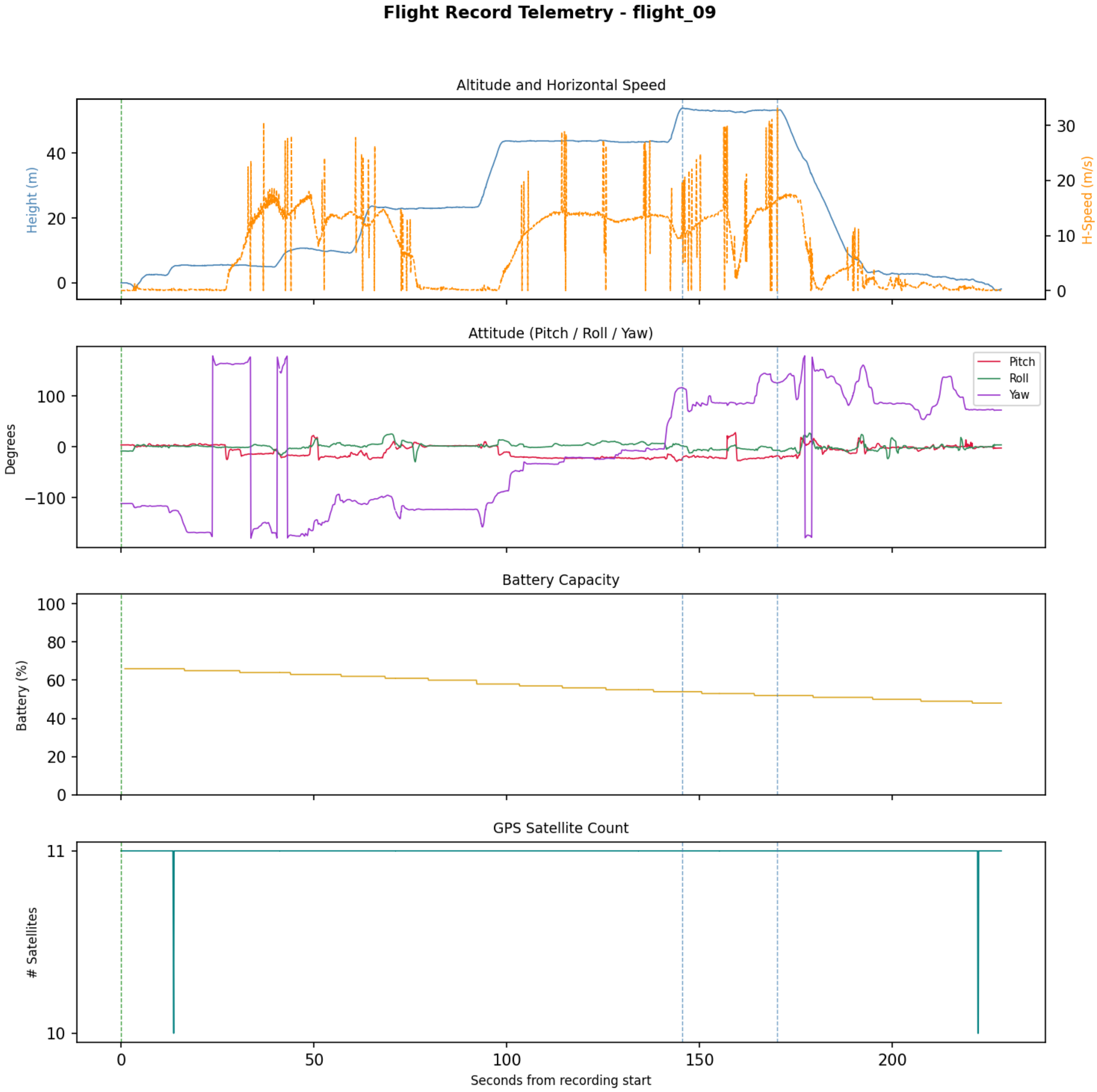
Timestamp (UTC)	Source	Event	Details
2018-06-20 15:03:42	drone_flight_storage:dl	Log started	P3S
2018-06-20 15:04:19	ctrl_and:fr (v3.1.38)	Log started	SN: 03Z1389682; App: 3.1.38; P3 Standard
2018-06-20 15:04:19	ctrl_and:fr (v3.1.38)	Motor turned on	on
2018-06-20 15:04:20	ctrl_and:fr (v3.1.38)	Record mode changed	No
2018-06-20 15:06:45	ctrl_and:fr (v3.1.38)	Reached peak height	53.9 m
2018-06-20 15:06:58	ctrl_and:fr (v3.1.38)	Record mode changed	Starting
2018-06-20 15:07:10	drone_flight_storage:dl	Reached peak height	55.8 m
2018-06-20 15:08:08	ctrl_and:fr (v3.1.38)	Log ended	Dist home: 1.3 m; Height: -1.9 m; Rec: 0.0 s
2018-06-20 15:08:37	drone_flight_storage:dl	Log ended	

Correlated Artefacts

Possibly correlated (indirect indicators):

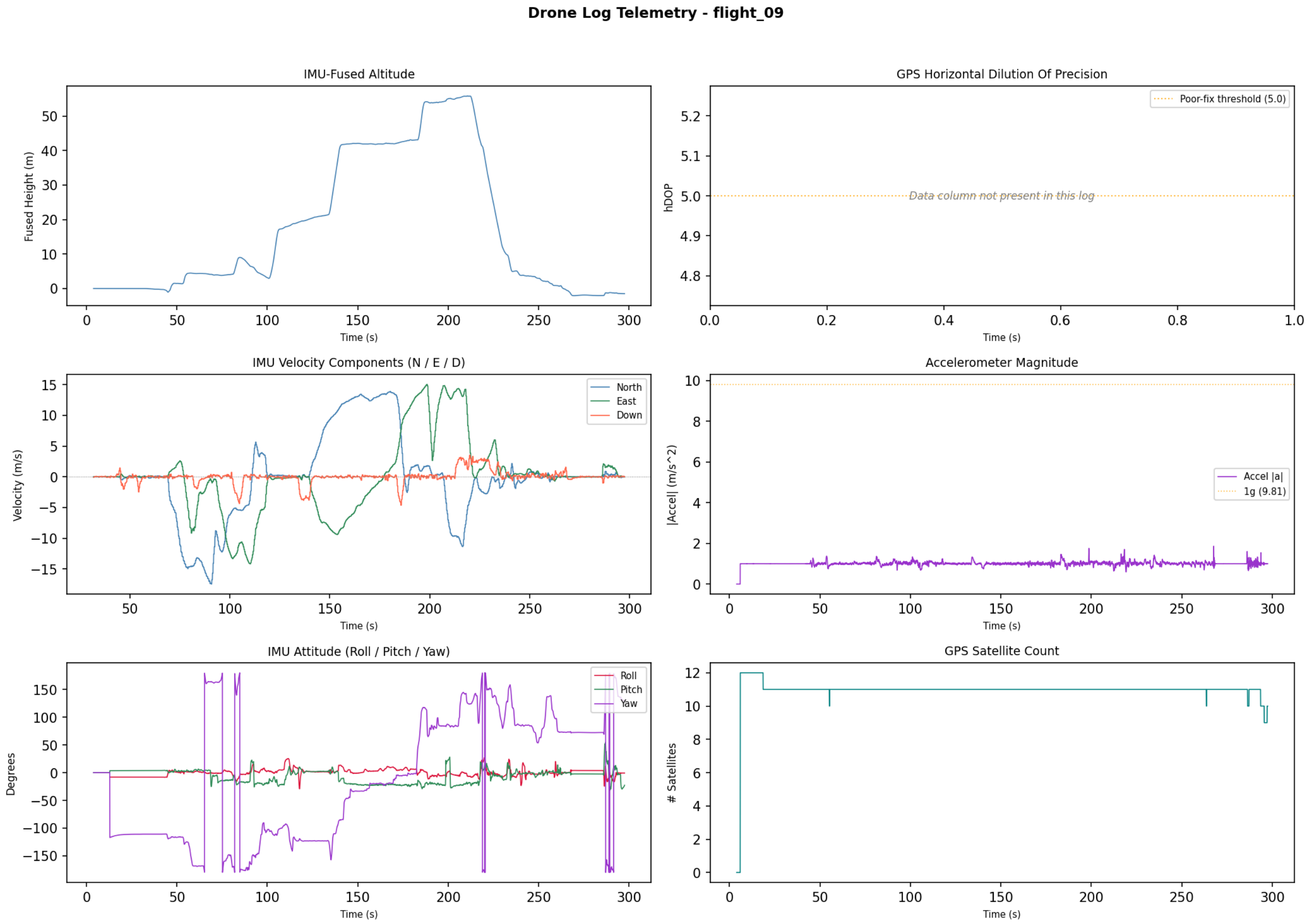
controller_android:videos	GPS within flight bounding box (lat 39.9623, lon -106.2198)
p5:fe28dc748339f681a0b0ec8fe9182f15b778ec23f27bf80d7998ddd6125961f7	

Flight Record - Sensor Telemetry



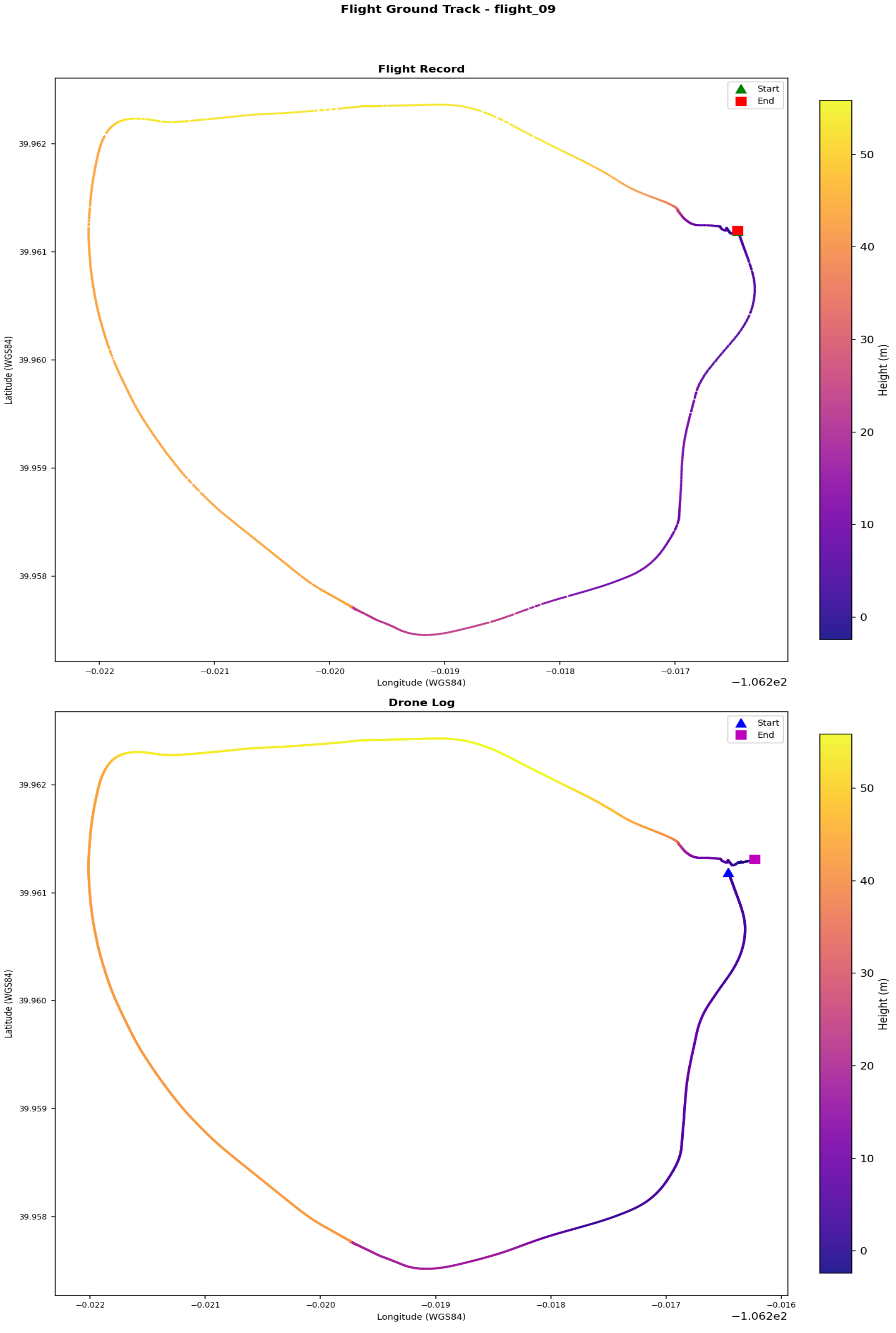
GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY013.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY013.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 10

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `34e6aa2645b4ec10935bcf1aa1efdd47cbcf9f3629c8d63b01e72d1f8dce6554`
Drone Log Name: FLY015.csv
Start: 2018-06-20T20:30:03.173000+00:00
End: 2018-06-20T20:39:23.185000+00:00
Duration: 560.0 s

Flight Metrics

Peak Height: 31.798286702144047 m
Distance Travelled: 775.5852552092165 m
Video Recording: No
Photo Capture: No

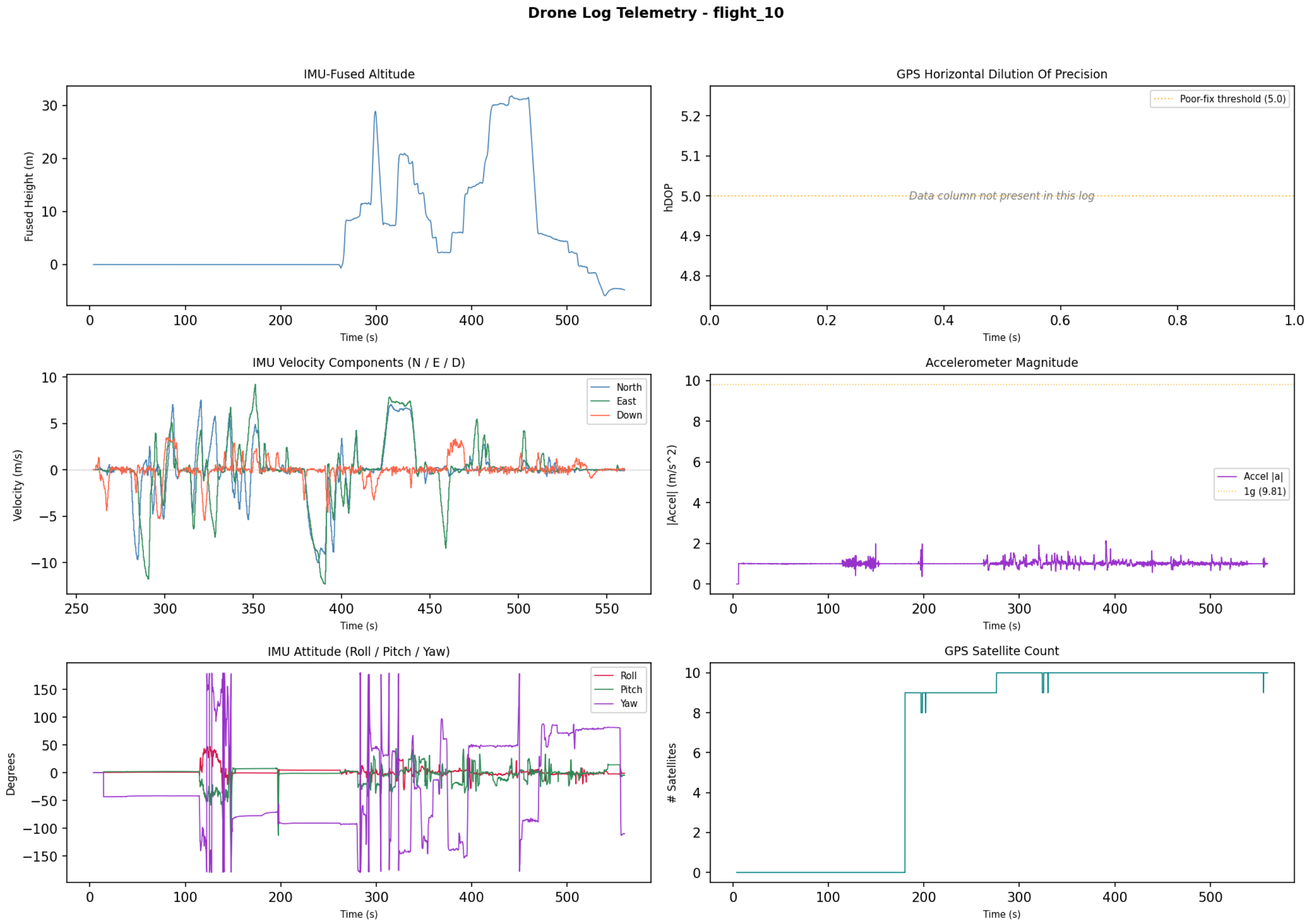
Fly Modes Observed (Chronological)

1. 2
2. 1
3. 3
4. 0

Key Events

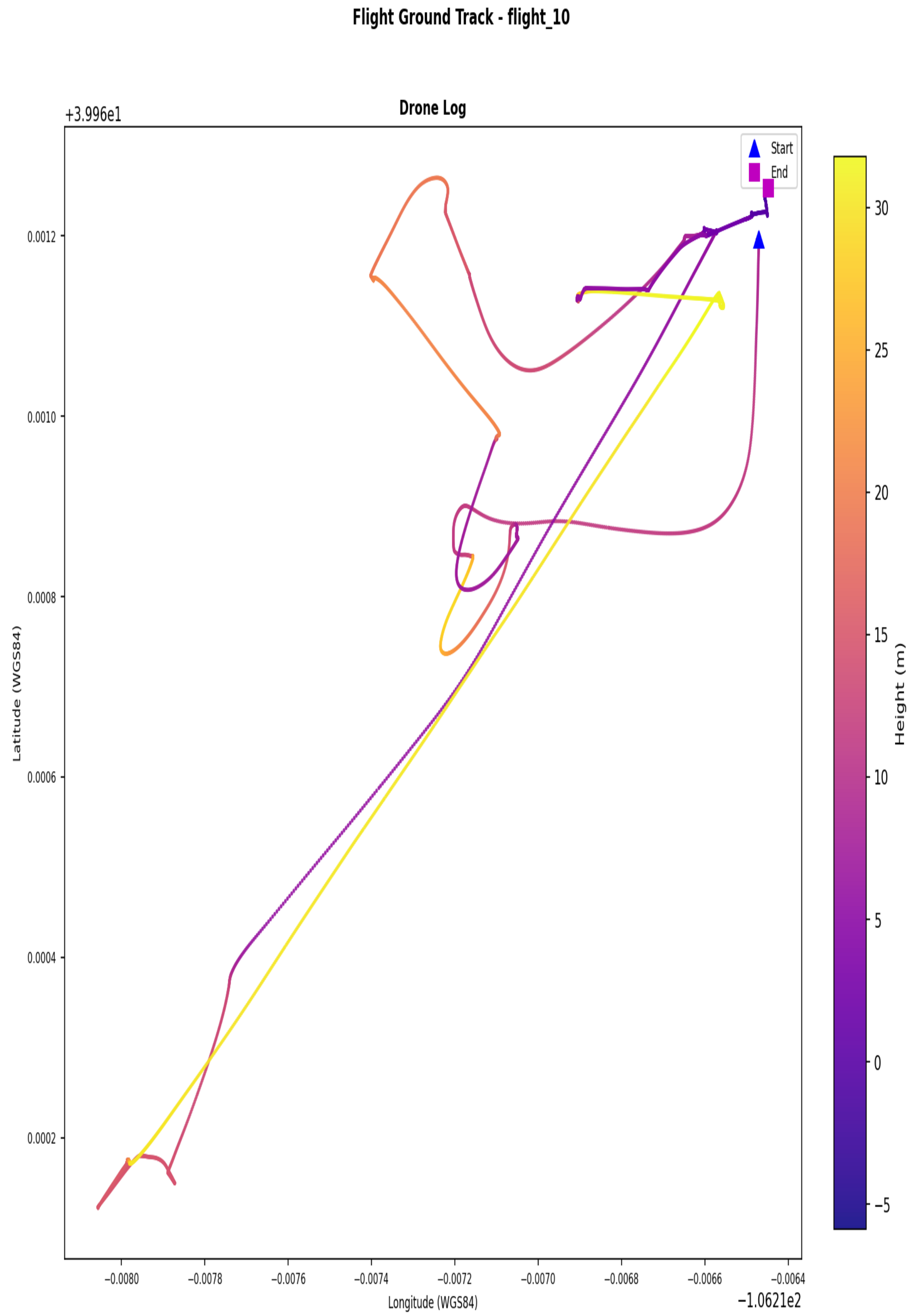
Timestamp (UTC)	Source	Event	Details
2018-06-20 20:32:17	drone_flight_storage:dl	Log started	P3S
2018-06-20 20:37:24	drone_flight_storage:dl	Reached peak height	31.8 m
2018-06-20 20:39:20	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY015.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).
A KML file (FLY015.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 11

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 4613ebd15162f415d0a68df86299496f34e43304bcf8651b0bb2b8f4e40c6ad5
Drone Log Name: FLY016.csv
Start: 2018-06-21T14:48:30.252000+00:00
End: 2018-06-21T14:50:25.420000+00:00
Duration: 115.2 s

Flight Metrics

Peak Height: 1.9128186593879792 m
Distance Travelled: 7.320542195992152 m
Video Recording: No
Photo Capture: No

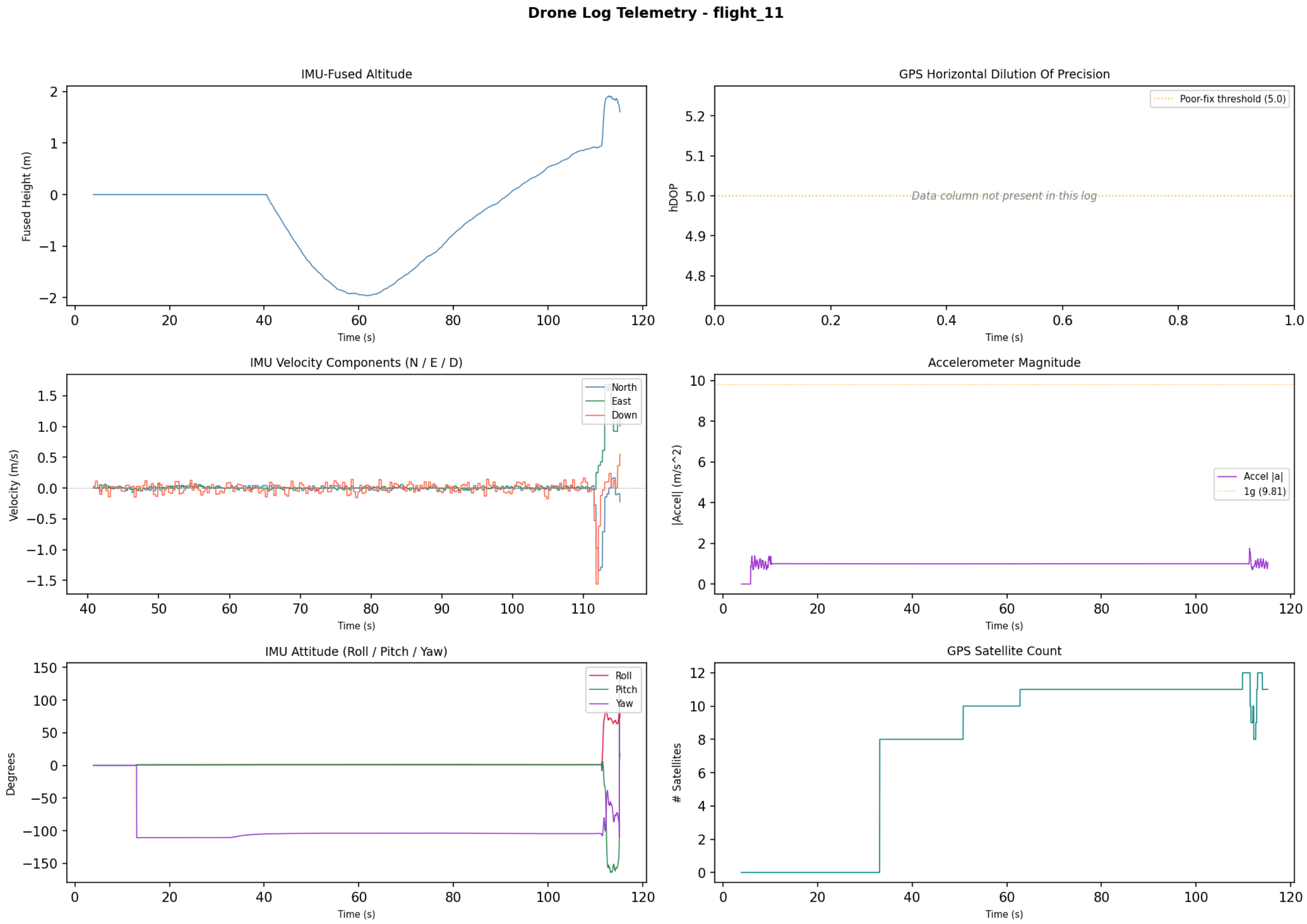
Fly Modes Observed (Chronological)

1. 0

Key Events

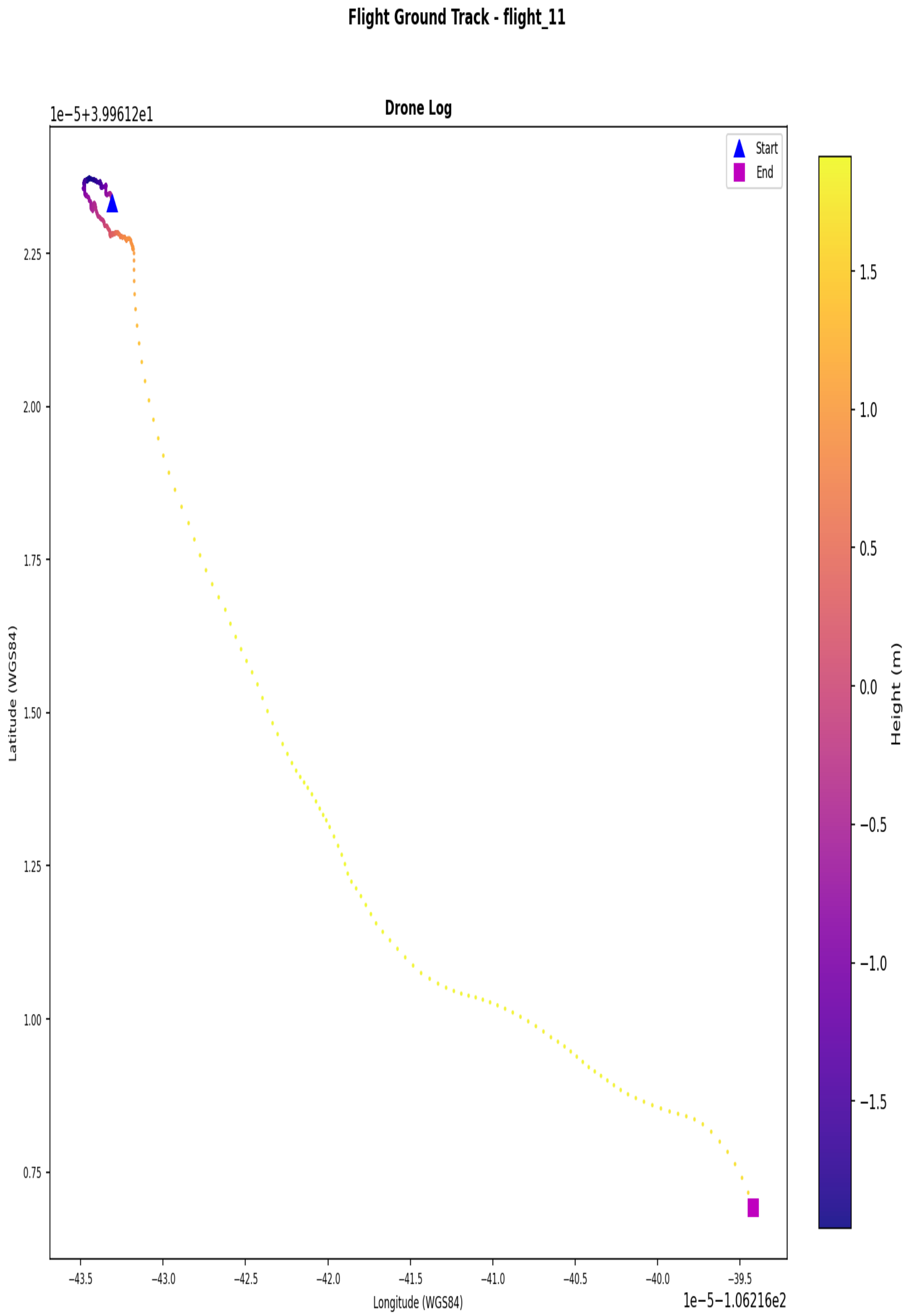
Timestamp (UTC)	Source	Event	Details
2018-06-21 14:48:47	drone_flight_storage:dl	Log started	P3S
2018-06-21 14:50:22	drone_flight_storage:dl	Reached peak height	1.9 m
2018-06-21 14:50:25	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY016.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY016.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 12

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `bfbb5b05396ef26b46229c5d5ca3ae235e5ae81589d2a96d8600ef6692aa2aad`
Drone Log Name: FLY017.csv
Start: 2018-06-21T15:51:40.662000+00:00
End: 2018-06-21T16:07:33.535000+00:00
Duration: 952.9 s

Flight Metrics

Peak Height: 107.54388886428954 m
Distance Travelled: 2933.0435180057566 m
Video Recording: No
Photo Capture: No

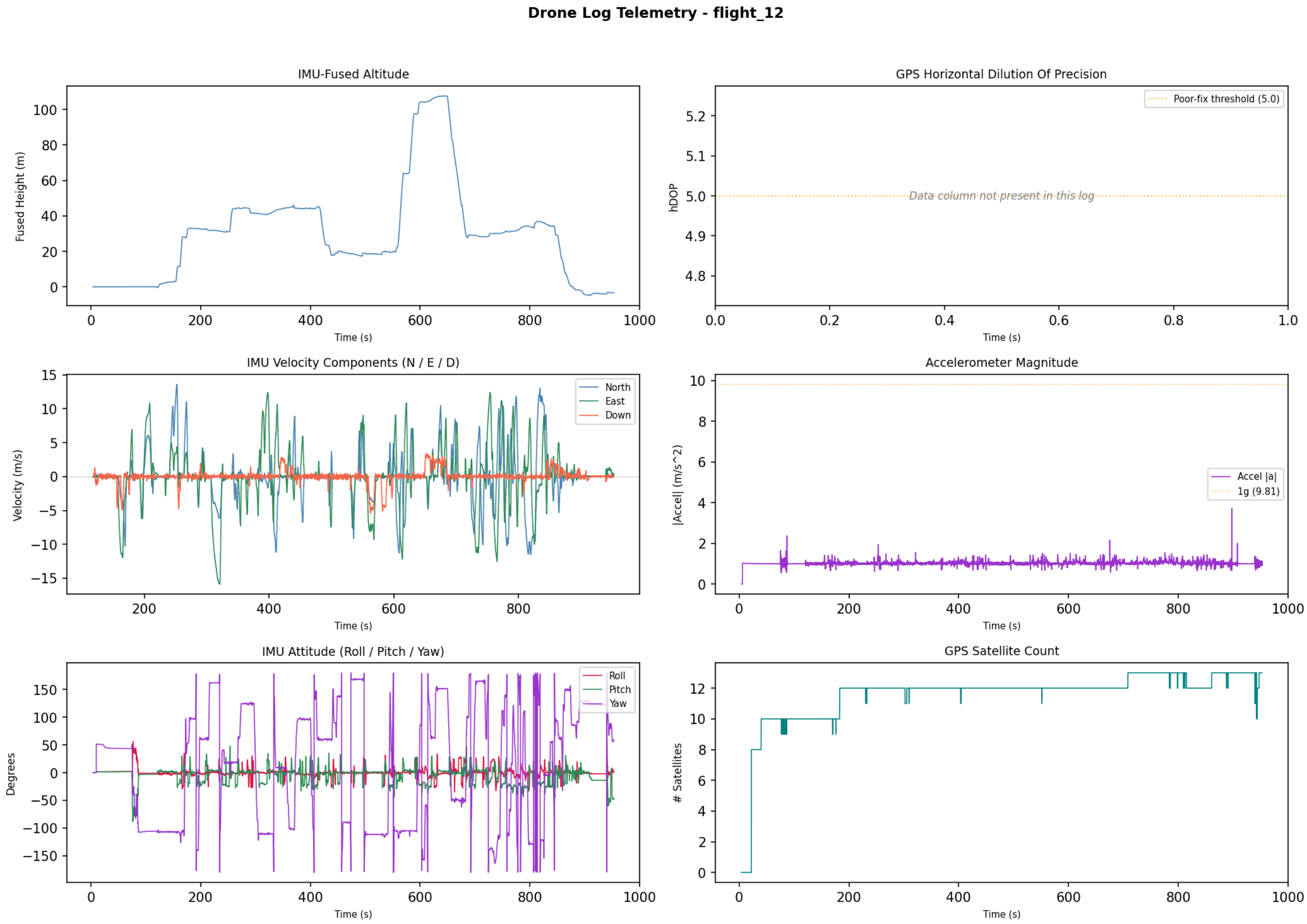
Fly Modes Observed (Chronological)

1. 2
2. 1
3. 3
4. 0

Key Events

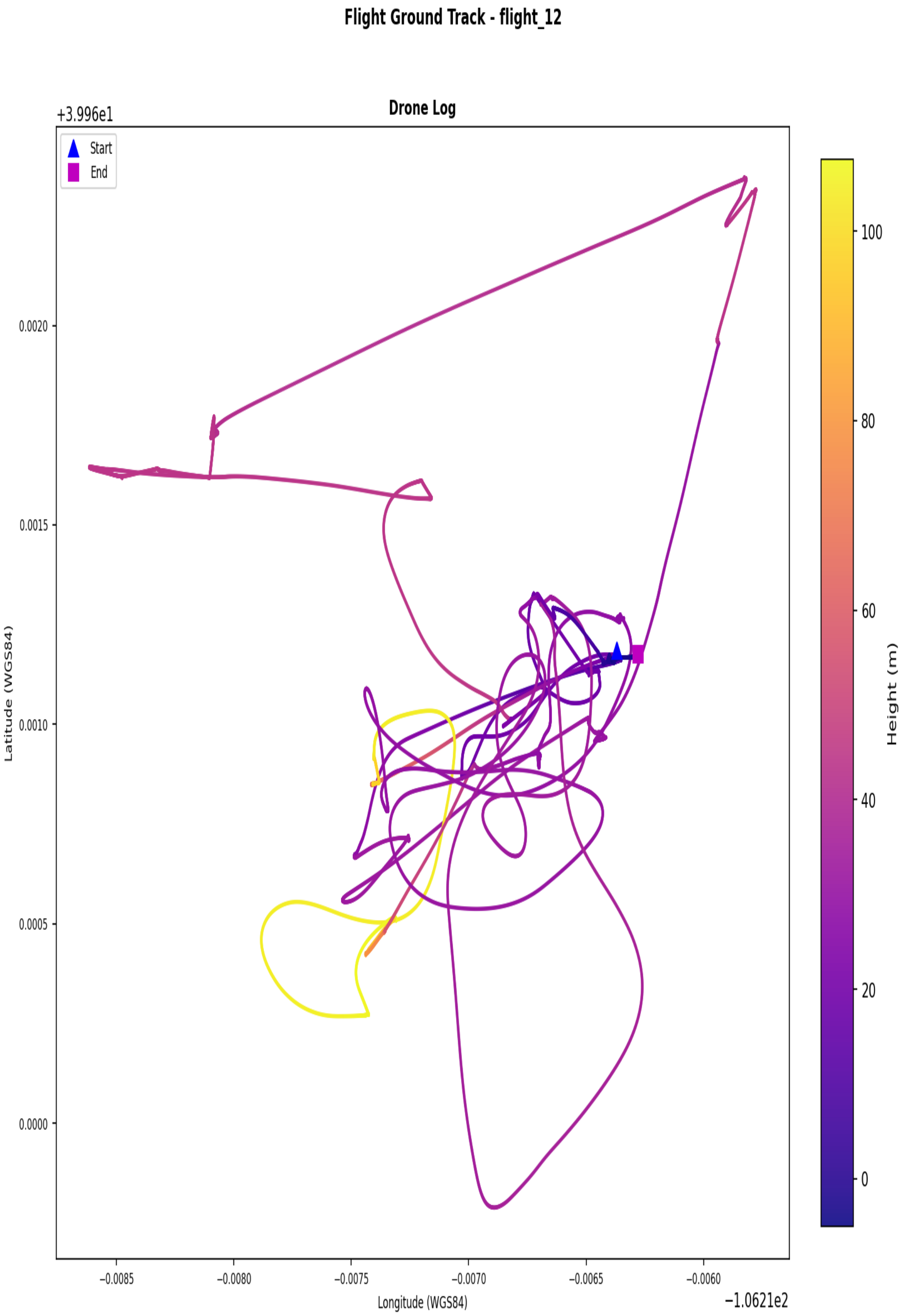
Timestamp (UTC)	Source	Event	Details
2018-06-21 15:51:52	drone_flight_storage:dl	Log started	P3S
2018-06-21 16:02:22	drone_flight_storage:dl	Reached peak height	107.5 m
2018-06-21 16:07:30	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY017.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).
A KML file (FLY017.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 13

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `483cbf7d95319254e6ec429a85882b9743bf93d202af4a08fa8d53691041b530`
Drone Log Name: FLY018.csv
Start: 2018-06-21T19:51:18.850000+00:00
End: 2018-06-21T19:52:57.875000+00:00
Duration: 99.0 s

Flight Metrics

Peak Height: 1.3970570316977093 m
Distance Travelled: 11381919.776069224 m
Video Recording: No
Photo Capture: No

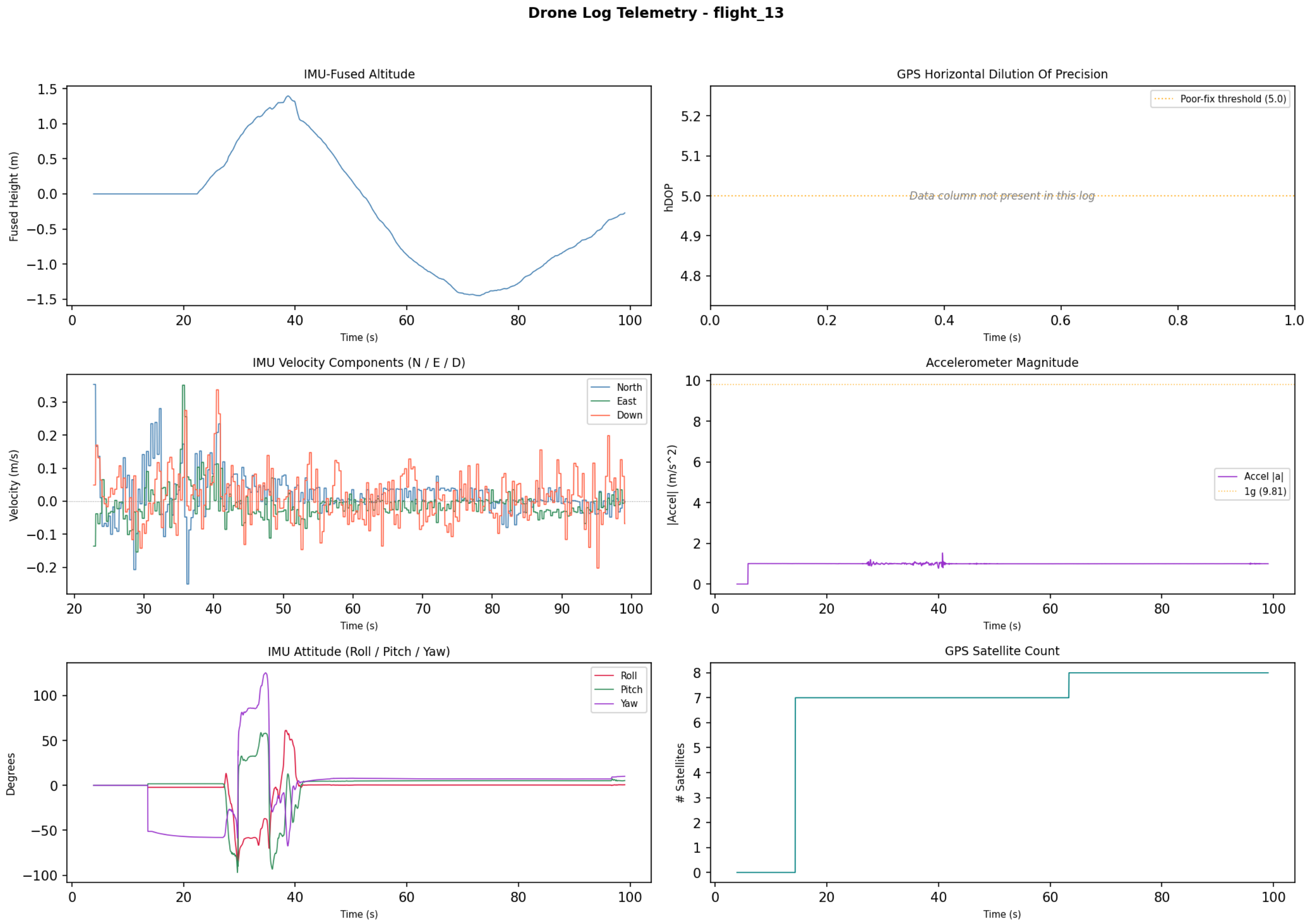
Fly Modes Observed (Chronological)

1. 0

Key Events

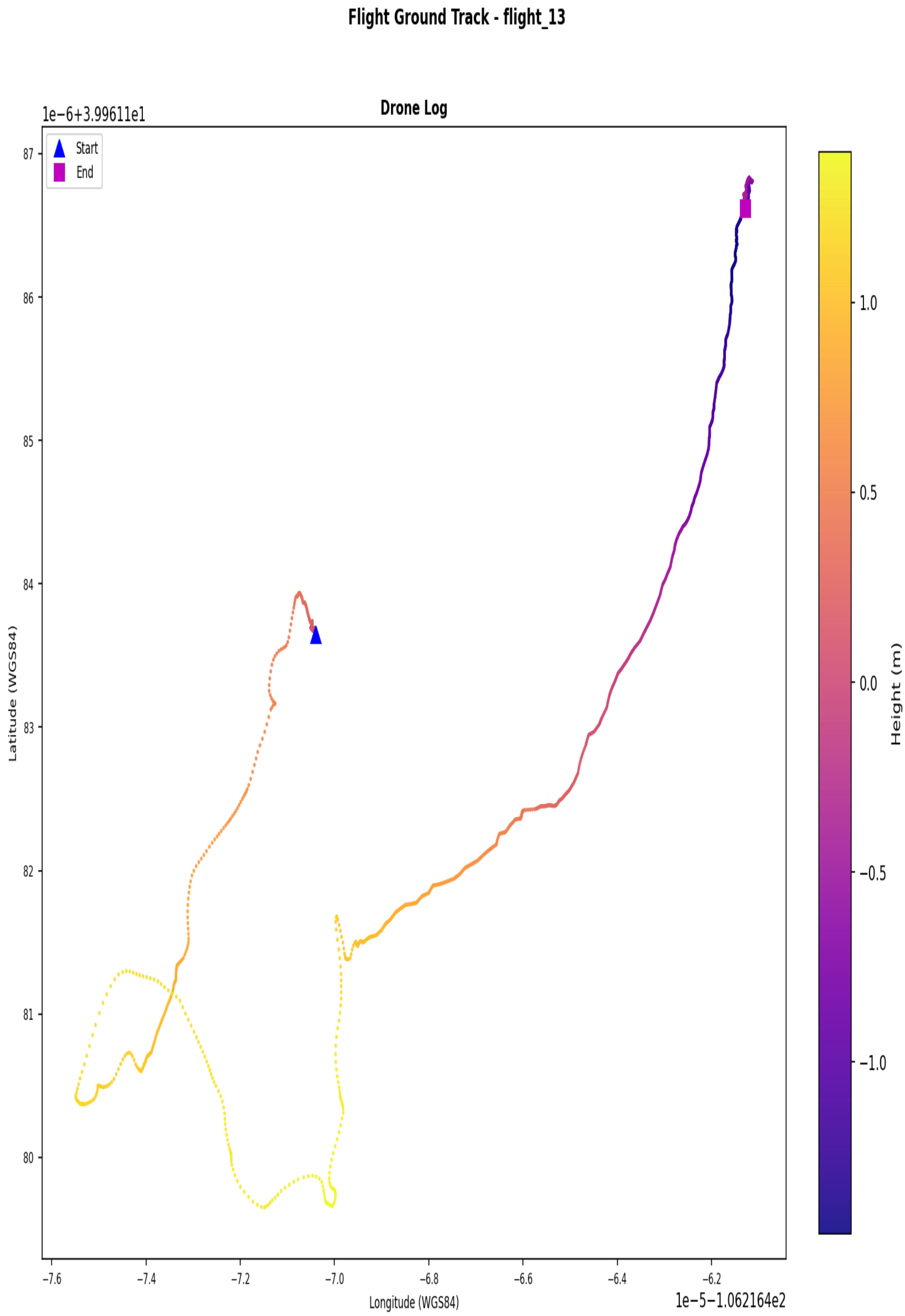
Timestamp (UTC)	Source	Event	Details
2018-06-21 19:51:47	drone_flight_storage:dl	Log started	P3S
2018-06-21 19:51:57	drone_flight_storage:dl	Reached peak height	1.4 m
2018-06-21 19:52:57	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY018.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).
A KML file (FLY018.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 14

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `9fbb6503b83f372741133cf21e5bd8c0d494755ccde59cf8fb721b334a5e85c5`
Drone Log Name: FLY019.csv
Start: 2018-06-21T19:56:18.117000+00:00
End: 2018-06-21T20:12:31.455000+00:00
Duration: 973.3 s

Flight Metrics

Peak Height: 8.298943365338397 m
Distance Travelled: 195.12062621097402 m
Video Recording: No
Photo Capture: No

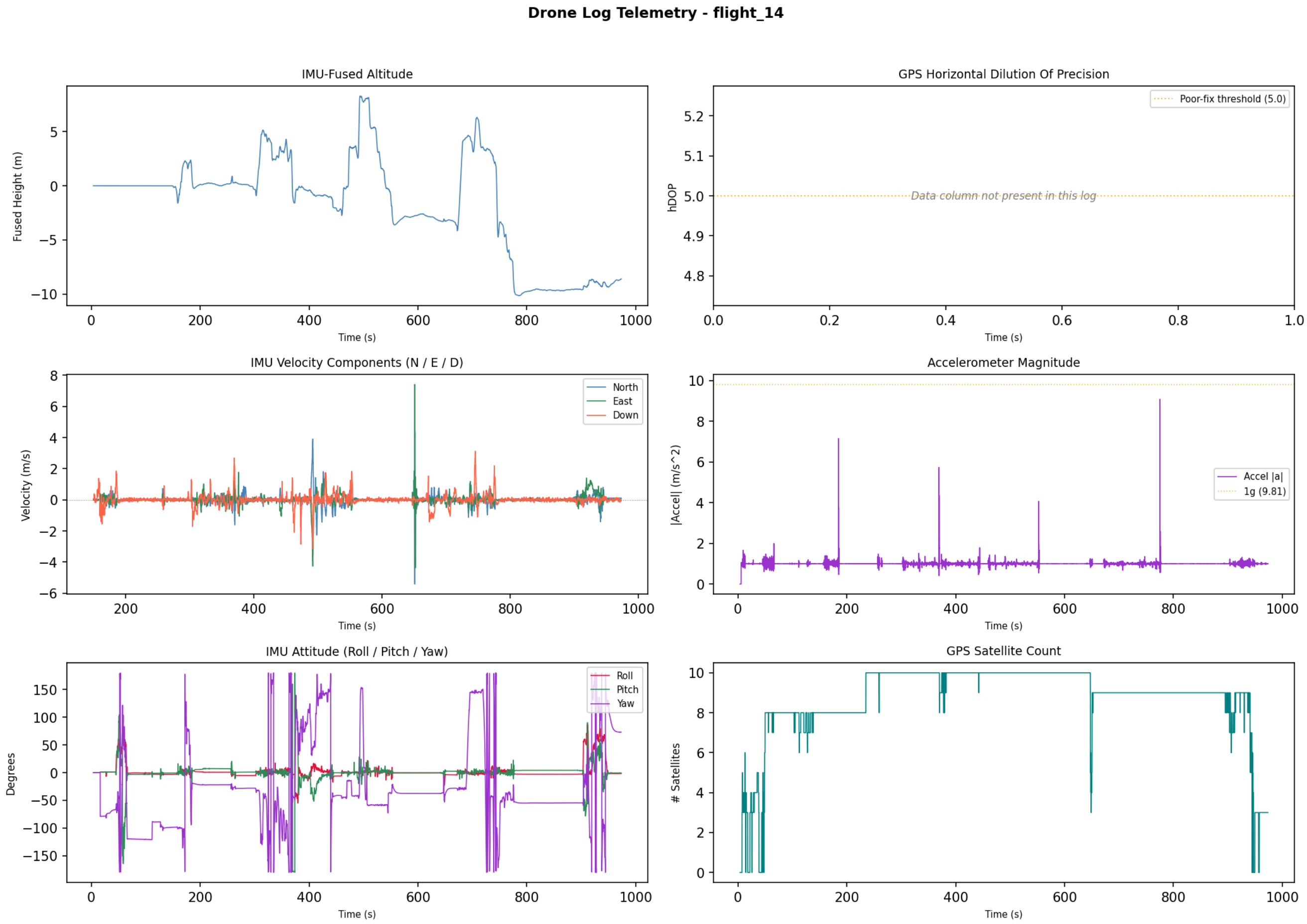
Fly Modes Observed (Chronological)

1. 0
2. 1
3. 2
4. 3

Key Events

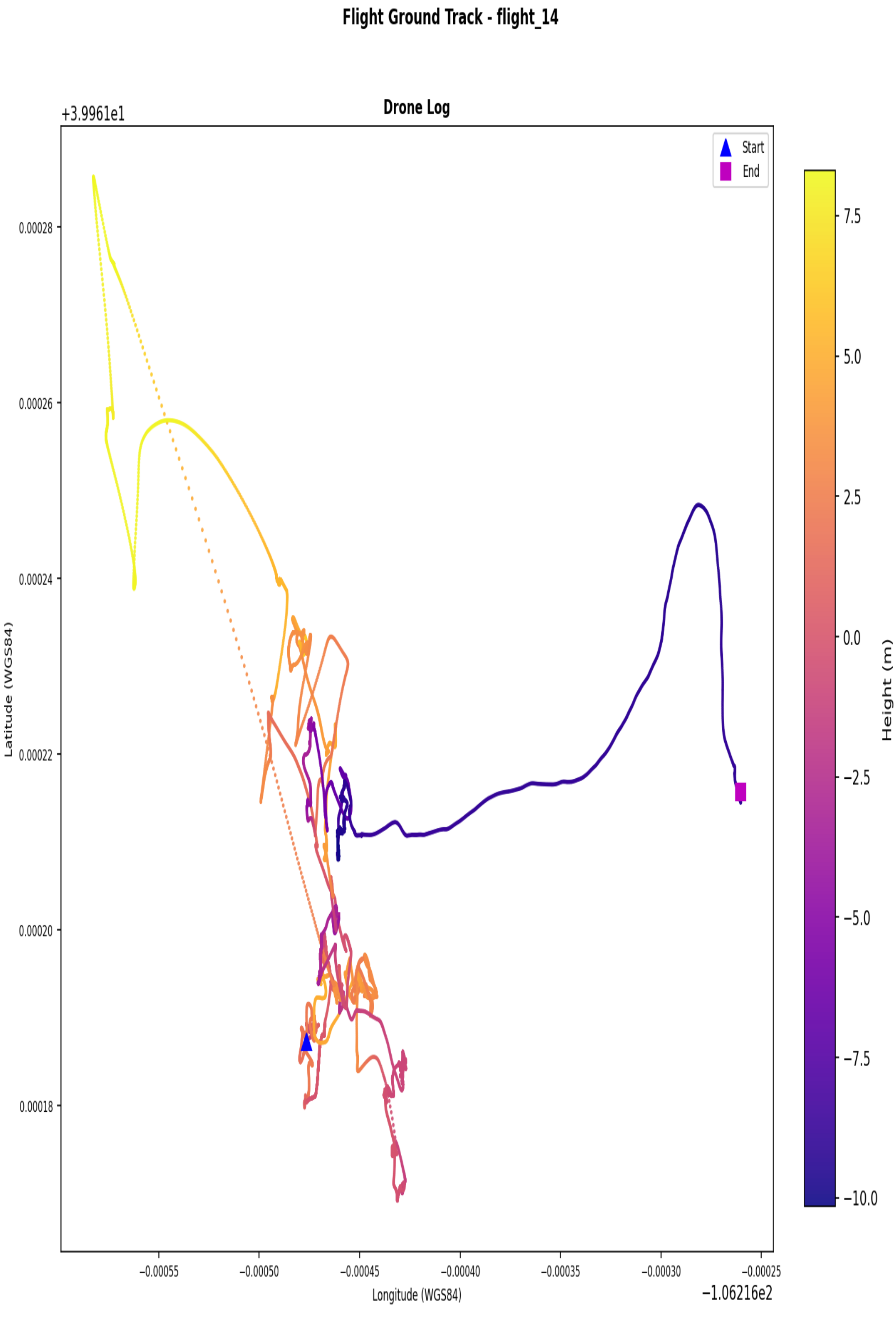
Timestamp (UTC)	Source	Event	Details
2018-06-21 19:56:22	drone_flight_storage:dl	Log started	P3S
2018-06-21 20:04:31	drone_flight_storage:dl	Reached peak height	8.3 m
2018-06-21 20:12:31	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY019.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY019.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 15

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `ce628b7ec2f92ed76df22d3c8b6fc73623375032b341358ef2a6adb619485767`
Drone Log Name: FLY020.csv
Start: 2018-06-21T20:12:38.107000+00:00
End: 2018-06-21T20:14:39.845000+00:00
Duration: 121.7 s

Flight Metrics

Peak Height: -
Distance Travelled: 7.562245330351284 m
Video Recording: No
Photo Capture: No

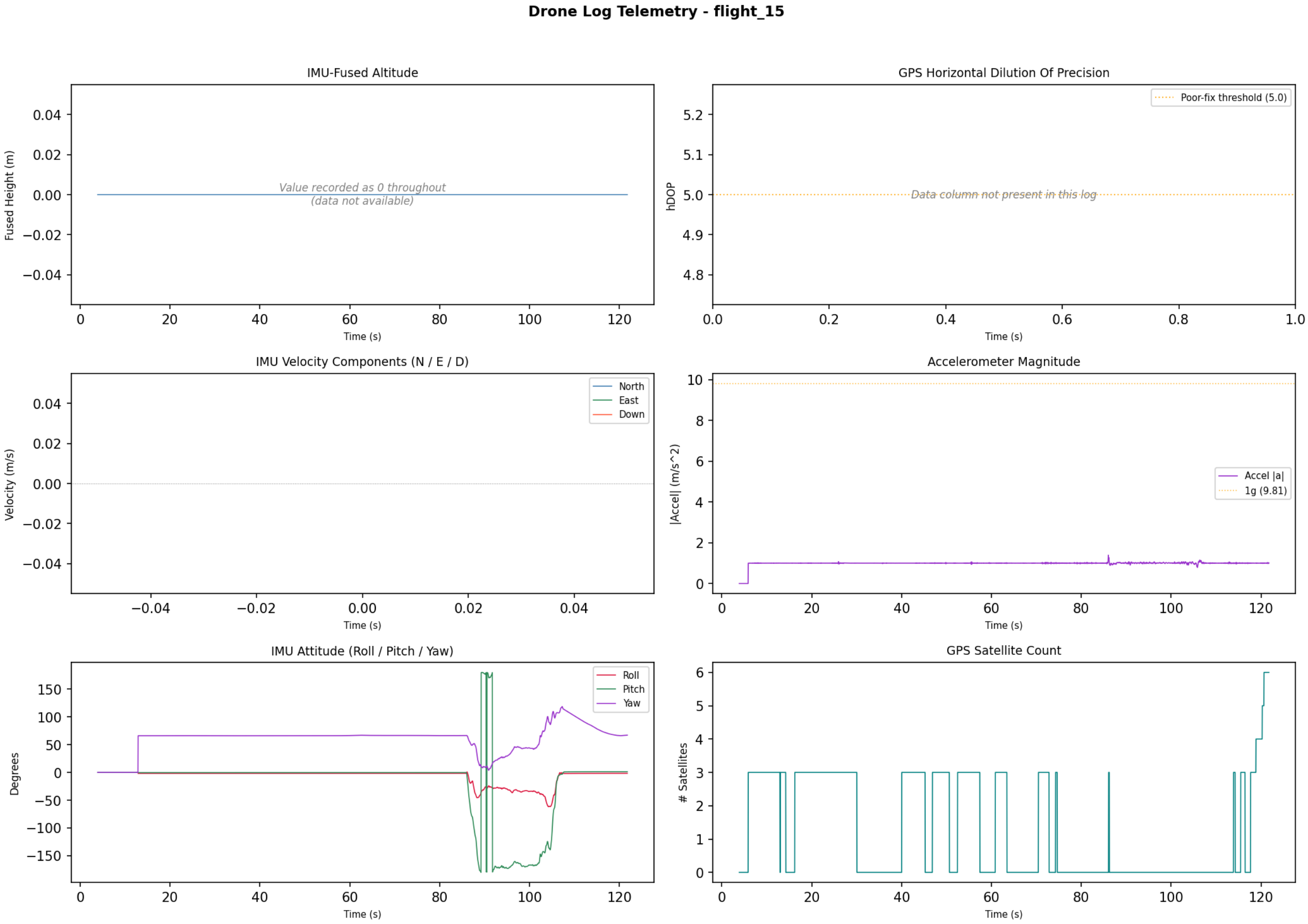
Fly Modes Observed (Chronological)

1. 0

Key Events

Timestamp (UTC)	Source	Event	Details
2018-06-21 20:12:42	drone_flight_storage:dl	Log started	P3S
2018-06-21 20:12:42	drone_flight_storage:dl	Reached peak height	-0.0 m
2018-06-21 20:14:43	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 16

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: `fac65daad99468629cc23b193aea2619860be905c8db00074f43536f950ef4d3`
Drone Log Name: FLY021.csv
Start: 2018-06-21T20:14:47.362000+00:00
End: 2018-06-21T20:15:40.185000+00:00
Duration: 52.8 s

Flight Metrics

Peak Height: -
Distance Travelled: -
Video Recording: No
Photo Capture: No

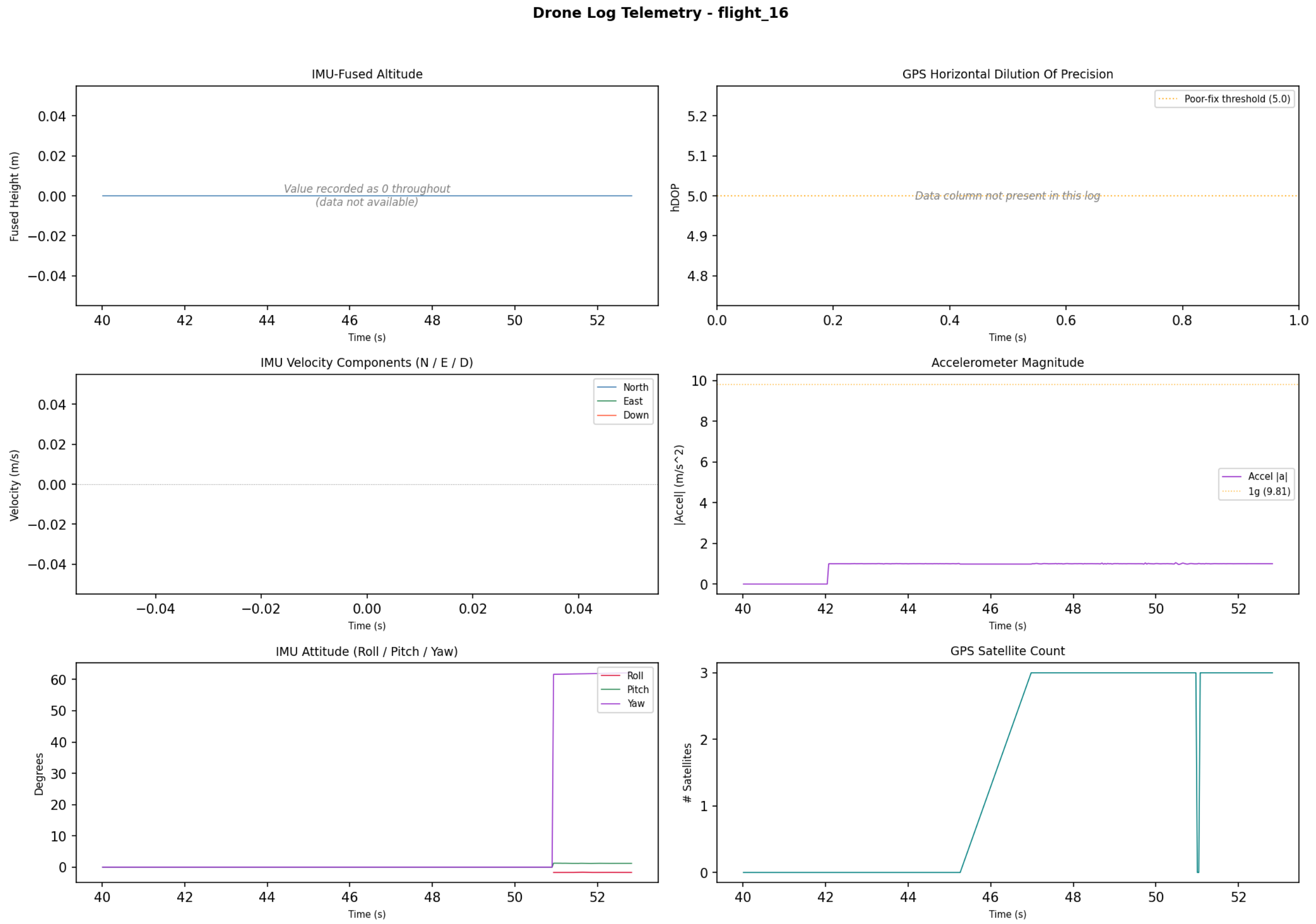
Fly Modes Observed (Chronological)

1. 0

Key Events

Timestamp (UTC)	Source	Event	Details
2018-06-21 20:14:48	drone_flight_storage:dl	Log started	P3S
2018-06-21 20:14:49	drone_flight_storage:dl	Reached peak height	-0.0 m
2018-06-21 20:15:40	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	Yes
controller_ios	No
drone_sd	No
drone_flight_storage	Yes

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_android	1	3	0	1	1	0	2
drone_flight_storage	0	0	22	0	0	0	0

Data Quality Notes per Category

Drone Logs (22 files):

- 22 file(s) with row anomalies
- 22 file(s) with empty columns
- 22 file(s) with constant-value columns

Flight Logs (1 files):

- formats: 1 heading_log

Flight Records (1 files):

- 1 file(s) with row anomalies
- 1 file(s) with empty columns
- 1 file(s) with constant-value columns

Videos (2 files):

- 1 file(s) contain(s) EXIF GPS coordinates
- 1 file(s) missing (normalisable) EXIF timestamp
- 1 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Drone_logs: 22 file(s) with row anomalies detected in telemetry data.
- Drone_logs: 22 file(s) with empty columns - some telemetry channels contain no data.
- Drone_logs: 22 file(s) with constant-value columns - some telemetry channels may be unreliable.
- Flight_logs: log files present in the following format(s): 1 heading_log.
- Flight_records: 1 file(s) with row anomalies detected in telemetry data.
- Flight_records: 1 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 1 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

1. [p2 - controller ios]: source evidence not found
2. [p3 - controller android - drone logs]: no artefacts found
3. [p3 - controller android - images]: no artefacts found
4. [p3 - controller ios]: source evidence not found
5. [p3 - drone sd]: source evidence not found
6. [p4 - controller ios]: source evidence not found
7. [p4 - drone sd]: source evidence not found
8. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY000.csv
9. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY001.csv
10. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY002.csv
11. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY003.csv
12. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY004.csv
13. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY014.csv

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Drone Logs

norm_FLY000.csv

- missing_gps: 47197 row(s) affected
- contains_no_value: 27 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+22 more)
- contains_constant_value: 62 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+57 more)
- quaternion_invalid: 86 row(s) affected

norm_FLY001.csv

- missing_gps: 8270 row(s) affected
- contains_no_value: 27 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+22 more)
- contains_constant_value: 63 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+58 more)

norm_FLY002.csv

- missing_gps: 8300 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+21 more)
- contains_constant_value: 62 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+57 more)
- quaternion_invalid: 257 row(s) affected

norm_FLY003.csv

- missing_gps: 679 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+21 more)

- contains_constant_value: 71 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+66 more)
- quaternion_invalid: 255 row(s) affected

norm_FLY004.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 3821 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+21 more)
- contains_constant_value: 65 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+60 more)
- quaternion_invalid: 228 row(s) affected

norm_FLY005.csv

- missing_gps: 7559 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+21 more)
- contains_constant_value: 57 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+52 more)
- quaternion_invalid: 838 row(s) affected

norm_FLY006.csv

- missing_gps: 3919 row(s) affected
- contains_no_value: 23 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+18 more)
- contains_constant_value: 48 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+43 more)
- quaternion_invalid: 257 row(s) affected

norm_FLY007.csv

- missing_gps: 3730 row(s) affected
- contains_no_value: 23 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+18 more)
- contains_constant_value: 48 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+43 more)
- quaternion_invalid: 257 row(s) affected

norm_FLY008.csv

- missing_gps: 41927 row(s) affected
- altitude_negative: 41 row(s) affected
- contains_no_value: 10 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+5 more)
- contains_constant_value: 14 column(s) - AirCraftCondition, AirCraftCondition:launch_state, Battery(0):maxVolts, GPS(0):numGLNAS, GPS(0):numGPS (+9 more)
- quaternion_invalid: 435 row(s) affected
- attitude_out_of_bounds: 3135 row(s) affected

norm_FLY009.csv

- missing_gps: 10166 row(s) affected
- contains_no_value: 12 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+7 more)
- contains_constant_value: 33 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+28 more)
- quaternion_invalid: 257 row(s) affected
- attitude_out_of_bounds: 90 row(s) affected

norm_FLY010.csv

- missing_gps: 245 row(s) affected
- contains_no_value: 34 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+29 more)
- contains_constant_value: 65 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+60 more)

- quaternion_invalid: 204 row(s) affected

norm_FLY011.csv

- missing_gps: 2515 row(s) affected
- contains_no_value: 12 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+7 more)
- contains_constant_value: 37 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+32 more)
- quaternion_invalid: 477 row(s) affected
- attitude_out_of_bounds: 13 row(s) affected

norm_FLY012.csv

- missing_gps: 3386 row(s) affected
- contains_no_value: 12 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+7 more)
- contains_constant_value: 32 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+27 more)
- quaternion_invalid: 257 row(s) affected

norm_FLY013.csv

- missing_gps: 8361 row(s) affected
- altitude_negative: 1247 row(s) affected
- contains_no_value: 10 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+5 more)
- contains_constant_value: 17 column(s) - AirCraftCondition, Battery(0):maxVolts, Battery(0):minCurrent, Battery(0):minWatts, GPS(0):Date (+12 more)
- quaternion_invalid: 257 row(s) affected

norm_FLY014.csv

- missing_gps: 297 row(s) affected
- contains_no_value: 32 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+27 more)
- contains_constant_value: 87 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+82 more)
- quaternion_invalid: 256 row(s) affected

norm_FLY015.csv

- missing_gps: 15807 row(s) affected
- altitude_negative: 1449 row(s) affected
- contains_no_value: 10 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+5 more)
- contains_constant_value: 17 column(s) - AirCraftCondition, AirCraftCondition:launch_state, Battery(0):maxVolts, Battery(0):minCurrent, Battery(0):minWatts (+12 more)
- quaternion_invalid: 300 row(s) affected
- attitude_out_of_bounds: 52 row(s) affected

norm_FLY016.csv

- missing_gps: 3176 row(s) affected
- altitude_negative: 1401 row(s) affected
- contains_no_value: 10 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+5 more)
- contains_constant_value: 41 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+36 more)
- quaternion_invalid: 262 row(s) affected
- attitude_out_of_bounds: 79 row(s) affected

norm_FLY017.csv

- missing_gps: 27096 row(s) affected
- altitude_negative: 2172 row(s) affected
- contains_no_value: 9 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+4 more)
- contains_constant_value: 16 column(s) - AirCraftCondition, Battery(0):maxVolts, Battery(0):minCurrent, Battery(0):minWatts,

GPS(0):numGLNAS (+11 more)
- quaternion_invalid: 170 row(s) affected

norm_FLY018.csv

- missing_gps: 2760 row(s) affected
- altitude_negative: 1315 row(s) affected
- contains_no_value: 10 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+5 more)
- contains_constant_value: 41 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+36 more)
- quaternion_invalid: 278 row(s) affected
- attitude_out_of_bounds: 10 row(s) affected

norm_FLY019.csv

- missing_gps: 27618 row(s) affected
- altitude_negative: 13890 row(s) affected
- contains_no_value: 10 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+5 more)
- contains_constant_value: 16 column(s) - AirCraftCondition, Battery(0):minCurrent, Battery(0):minWatts, GPS(0):Date, GPS(0):numGLNAS (+11 more)
- quaternion_invalid: 357 row(s) affected
- attitude_out_of_bounds: 418 row(s) affected

norm_FLY020.csv

- missing_gps: 3361 row(s) affected
- contains_no_value: 24 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+19 more)
- contains_constant_value: 41 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+36 more)
- quaternion_invalid: 257 row(s) affected
- attitude_out_of_bounds: 505 row(s) affected

norm_FLY021.csv

- missing_gps: 365 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV1, General:absoluteHeight:C, General:absoluteHeight:C, General:relativeHeight:C (+21 more)
- contains_constant_value: 59 column(s) - AirCraftCondition, AirCraftCondition:cos_safe_tilt, AirCraftCondition:craft_flight_mode, AirCraftCondition:dynamic_thrust, AirCraftCondition:hover_thrust (+54 more)
- quaternion_invalid: 264 row(s) affected

Flight Logs

20-06-2018

- format: heading_log | entries: 17

Flight Records

norm_DJIFlightRecord_2018-06-20_[09-04-19].csv

- altitude_negative: 69 row(s) affected
- contains_no_value: 37 column(s) - APP_SER_WARN.warn, CENTER_BATTERY.isBatteryOnCharge, CENTER_BATTERY.isNeedStudy, CENTER_BATTERY.lastStudyCycle, CENTER_BATTERY.totalStudyCycle (+32 more)
- contains_constant_value: 174 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, APP_TIP.tip, CAMERA_INFO.cameraType (+169 more)

Videos

2018_06_20_09_06_58.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Note: no controller device name found - Device and account information.
2. Drone Model inconsistency detected - conflicting values observed: P3 Standard, P3S. Manual verification required.
3. Note: no controller device found - Device and account information.
4. Note: no account e-mail found - Device and account information.
5. Note: no last country code found - Device and account information.
6. Note: no aircraft model code found - Device and account information.
7. Note: no controller serial number found - Device and account information.
8. Note: no account nickname found - Device and account information.
9. 15 flight record(s) could not be correlated to a drone log (flight_01, flight_02, flight_03, flight_04, flight_05, flight_06, flight_07, flight_08, flight_10, flight_11, flight_12, flight_13, flight_14, flight_15, flight_16). Independent corroboration is absent for these flight(s).
10. 3 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

dji.pilot.xml	Account Data	controller_android	extract_logical
SHA-256:	84f53d609fdcba690fe2e45150e538659e99f29b93a7ca004c9579a3dd7af3d7		
SHA-1:	0e3ee1db96c80ce96700d8aaa4eaeb49ff625f9a		

dji.db	Databases	controller_android	extract_logical
SHA-256:	edfa3a9ff4f682f022d3bb0cd51bda350a1928e47b67ea8a237f72dd25247510		
SHA-1:	5d67a0a990b1492ae2db3eaa407533e81228904e		

flysafe_areas_djigo.db	Databases	controller_android	extract_logical
SHA-256:	460e673d983eb15bcce465a60ee104b080e8b0d8769731445cc342214ff2b3ba		
SHA-1:	421a149f3722a55ac81a97324d75b7bd1d1a95a7		

logdb.db	Databases	controller_android	extract_logical
SHA-256:	d3acb5821a77cb0213739265b08f75ed4557ed4c84922763124a4db6bbc27839		
SHA-1:	bafe0b87adf37c679bd628a93044ce1aa6912949		

20-06-2018	Flight Logs	controller_android	extract_logical
SHA-256:	3b279efed7db98ca8bea2e904cf88d82ba3ee5cc013112a658ebbf07ef8ba49f		
SHA-1:	8469bf39e5ce1ad075f35f1ea754f27639318665		

DJIFlightRecord_2018-06-20_[09-04-19].txt	Flight Records	controller_android	extract_logical
SHA-256:	8815b0530e61b5b3c7d7ac1b5d526184e4bcf1e7911bb280c0fa72c9da1440e4		
SHA-1:	14f91065e4db9ff8c649252257cf9f8eb76c0535		

2018_06_20_09_06_58.info	Videos	controller_android	extract_logical
SHA-256:	fe28dc748339f681a0b0ec8fe9182f15b778ec23f27bf80d7998ddd6125961f7		
SHA-1:	65befc140b4a0eee7f405acd124fb948313814a7		

2018_06_20_09_06_58.mp4	Videos	controller_android	extract_logical
SHA-256:	0bacf331a1916bb1a0295977549083329fac4f3918120d98069dd2de50cefd1e		
SHA-1:	1ba5ced51900ef2dabb37108f015723c075fc9fa		

FLY000.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	4ef04116a2601f795d11d13e37924f7809d77f77255178c9fc4d4ead74073e81		
SHA-1:	03a1448280db64f7228dfeda55cfd1264bdd18db		

FLY001.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	29734f5dbafcbd7675a3d299f27b382045fcc3bef95e3afadc3f74da7dfe60dc		
SHA-1:	a3abce32ab37fb6a2bdcecd4a0c26268f0bb6e02		

FLY002.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	0de262d6ef03056c48e5ae387a46faa6ff0c62e3080738ec49373f47d39e83d8		
SHA-1:	4d4627b4211dc07ebce661c73c46d87d534e9857		

FLY003.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	d41201c54dd0309c910c0f1d0dc8520bbdd65bc2589a3a521ab3c534b039fcbe		
SHA-1:	75e39760401d2d1208be3b30554e1d9ecc627f08		

FLY004.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	87431f5861ce4cec730239a79332c416d98692ecda932c6d8be86f12888678d2		
SHA-1:	c3d16f9f1ba262feb4c20252fc6d7d1b7510cea2		

FLY005.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	e9db822cb9c76eb64676283c6b60640065b923e345dec124714c222f5f93609f		
SHA-1:	54c4d5bfa7d1cd571ea143acd821bde41b71582a		

FLY006.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	9ea14e35558b4e433c255e3e6bfd8dcf9c7f483e924f377194c3a5d984ecf384		
SHA-1:	dde1b2ba0cb1666ddc1af6b8d58b799516e7a681		

FLY007.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	4c770abc022a1797f7d1eef4182fa2bead4bb6663f5f1ef0e0b2a9903666a0f8		
SHA-1:	823dbe0b5bad6417616180f8cbe26d778f70ed42		

FLY008.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	7831a998afb0ee5203ec560962c0522a91e06087b005640846c72265e9ece228		
SHA-1:	fa7894eefcffb28af65d0f25461e3e6ae76197a4		

FLY009.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	33857c854395cf6ef00e5511a7d361bf8d30942a47040d1194e09fe466aafc2e		
SHA-1:	059475c3e4ffd60ca65004b7647dc5841d865973		

FLY010.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	b398ea030b651202f96d68179de4090a55499734d974acd61ec331fc1cacab43		
SHA-1:	b2092f0dd395fbfa028d0c64043514ea1d70361c		

FLY011.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	67e78ae79cdb21d4fae9c4a42eb00c89852b827790d134dde4c1c49b11c921b2		
SHA-1:	17e95b242784290c5481c6bf41ec06175d3eeed6		

FLY012.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	0056b3201a3675a1f2bdf6828458d0b7d6ce40f6176c70713b5e2129c77e1f17		
SHA-1:	ad42dd7196d12f374b0e4f5bcf30b9453a933df1		

FLY013.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	ca5d32c4fcb6a727bd992e3cc10a3a75dcd2b45aa6e80d9c1c12a54894bfed0b		
SHA-1:	e3a187a48a3aee3e819dc295e043153579cefed9		

FLY014.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	56cdd28dbb2cd7949778d26aea0e9f62e49ebe51752ddaac6609af84b3d5f216		
SHA-1:	23594187b7df1d08e7964ba42c2071da62d5bd57		

FLY015.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	ec937557175b7c83d6e57ced8fad1a8241210e5c1aefc34a646165d9f7c12fbc		
SHA-1:	1d33bcd18e4ef8ea010f4f6893f327c3b235550d		

FLY016.DAT	Drone Logs	drone_flight_storage	extract_logical
SHA-256:	b29e39bcfeb94274938e4670c541763b2450e5ef58ffd242056e4266e2d5d46e		
SHA-1:	45749aa63727ba77b38d6efb79dc8712a4b99ed7		

FLY017.DAT		Drone Logs	drone_flight_storage	extract_logical
SHA-256:	858ba001e4c0b0fb0328ffbaa1a6f6bab3427244014d6f3769a742e3e96739cc			
SHA-1:	65403458212449c56983d896f0ea8df0042cce05			

FLY018.DAT		Drone Logs	drone_flight_storage	extract_logical
SHA-256:	0854ca8d4571be8af0397baf76382d030f34b696129f0691f794ed8be7b5a8a7			
SHA-1:	62d36bdef3c715248bb657d0732f8733cceb15b6			

FLY019.DAT		Drone Logs	drone_flight_storage	extract_logical
SHA-256:	f7445b78f56dd84f6d349b3b0ec7c0e32a198bf88cc796e5f6a9f145bf00940f			
SHA-1:	2643fa811601104309aacee1300036409cb0e784			

FLY020.DAT		Drone Logs	drone_flight_storage	extract_logical
SHA-256:	96cf3185397ebd447880f6e184fc6429561032d6d60eb8e7fe821dcc11a0983d			
SHA-1:	ebf6d1046c7f4c4c6beeeff4470756bb2d011f44			

FLY021.DAT		Drone Logs	drone_flight_storage	extract_logical
SHA-256:	2bd4275ffb9ee3138fb2c1f94ac14694f32fb1821b4a19b251f8c57776e3396f			
SHA-1:	0aefd07c1306aacdd0557582945874802424ec13			

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Android

Category	P3 Extracted	P4 Decoded	P5 Normalised
Account Data	1	1	0
Databases	3	0	0
Flight Logs	1	0	0
Flight Records	1	1	1
Videos	2	2	0

Drone Flight Storage

Category	P3 Extracted	P4 Decoded	P5 Normalised
Drone Logs	22	44	22

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-15 15:07:10 parser_android v- rc=n/a

cmd: android_logical.zip C:\Users\Floris\Documents\dfdof_output\CASE-df001-2018-android\p2_image_parsing\controller_android_parsed

out: controller_android_parsed

2026-06-15 15:07:30 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2018-06-20_[09-04-19].txt DJIFlightRecord_2018-06-20_[09-04-19].csv

out: DJIFlightRecord_2018-06-20_[09-04-19].csv

2026-06-15 15:07:31 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2018_06_20_09_06_58.mp4

out: 2018_06_20_09_06_58_exif.json

2026-06-15 15:12:22 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY000.DAT

out: FLY000.csv, FLY000.kml

2026-06-15 15:12:23 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY001.DAT

out: FLY001.csv, FLY001.kml

2026-06-15 15:12:24 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY002.DAT

out: FLY002.csv, FLY002.kml

2026-06-15 15:12:25 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY003.DAT

out: FLY003.csv, FLY003.kml

2026-06-15 15:12:25 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY004.DAT

out: FLY004.csv, FLY004.kml

2026-06-15 15:12:26 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY005.DAT

out: FLY005.csv, FLY005.kml

2026-06-15 15:12:27 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY006.DAT

out: FLY006.csv, FLY006.kml

2026-06-15 15:12:27 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY007.DAT

out: FLY007.csv, FLY007.kml

2026-06-15 15:12:28 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY008.DAT

out: FLY008.csv, FLY008.kml

2026-06-15 15:12:29 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY009.DAT

out: FLY009.csv, FLY009.kml

2026-06-15 15:12:29 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY010.DAT

out: FLY010.csv, FLY010.kml

2026-06-15 15:12:30 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY011.DAT

out: FLY011.csv, FLY011.kml

2026-06-15 15:12:31 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY012.DAT

out: FLY012.csv, FLY012.kml

2026-06-15 15:13:06 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY013.DAT

out: FLY013.csv, FLY013.kml

2026-06-15 15:13:09 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY014.DAT

out: FLY014.csv, FLY014.kml

2026-06-15 15:13:09 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY015.DAT

out: FLY015.csv, FLY015.kml

2026-06-15 15:13:10 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY016.DAT

out: FLY016.csv, FLY016.kml

2026-06-15 15:13:11 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY017.DAT

out: FLY017.csv, FLY017.kml

2026-06-15 15:13:12 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY018.DAT

out: FLY018.csv, FLY018.kml

2026-06-15 15:13:13 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY019.DAT

out: FLY019.csv, FLY019.kml

2026-06-15 15:13:13 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY020.DAT

out: FLY020.csv, FLY020.kml

2026-06-15 15:13:14 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY021.DAT

out: FLY021.csv, FLY021.kml

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap >= 60 s AND >= 90% of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs <= 25 m with >= 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a +-2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.